

**Nonlinear wave interactions in rotating
shallow water equations on the sphere**

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THESIS PRESENTED TO THE
INSTITUTE OF MATHEMATICS, STATISTICS
AND COMPUTER SCIENCE
OF THE UNIVERSITY OF SÃO PAULO
IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS
FOR THE DEGREE OF
MASTER OF SCIENCE

Program: Applied Mathematics

Advisor: Prof. Dr. Pedro da Silva Peixoto

Coadvisor: Prof. Dr. Breno Raphaldini Ferreira da Silva

During the development of this work, the author received
financial support from FAPESP – grant 2024/01000-9

São Paulo
November, 2025

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This version of the thesis includes the corrections and modifications suggested by the Examining Committee during the defense of the original version of the work, which took place on November 10, 2025.

A copy of the original version is available at the Institute of Mathematics and Statistics of the University of São Paulo.

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Ficha catalográfica elaborada com dados inseridos pelo(a) autor(a)
Biblioteca Carlos Benjamin de Lyra
Instituto de Matemática e Estatística
Universidade de São Paulo

Dourado, Marco Antonio

Nonlinear wave interactions in rotating shallow water
equations on the sphere / Marco Antonio Dourado; orientador,
Pedro da Silva Peixoto; coorientador, Breno Raphaldini
Ferreira da Silva. – São Paulo, 2025.

90 p.: il.

Dissertação (Mestrado) – Programa de Pós-Graduação em
Matemática Aplicada / Instituto de Matemática e Estatística
/ Universidade de São Paulo.

Bibliografia

Versão original

1. Nonlinear interactions. 2. Shallow Water. 3.
gravity waves. 4. Rossby-Haurwitz waves. I. Peixoto,
Pedro da Silva. II. Título.

Bibliotecárias do Serviço de Informação e Biblioteca
Carlos Benjamin de Lyra do IME-USP, responsáveis pela
estrutura de catalogação da publicação de acordo com a AACR2:
Maria Lúcia Ribeiro CRB-8/2766; Stela do Nascimento Madruga CRB 8/7534.

*Ao meu querido e amado avô, João Lima
Dourado. Et lux perpetua luceat eis*

Agradecimentos

Aos meus pais Ana e Marcelo, à minha irmã Maria e minha família, por todo carinho e apoio ao longo de toda a vida.

À minha namorada Giovanna, por seu amor, e a toda sua família.

Aos meus orientadores Pedro e Breno, por todo o suporte, empenho, dedicação, paciência e boas conversas durante o mestrado.

Aos professores André Teruya e Carlos Raupp, por toda a ajuda concedida durante o projeto.

Aos meus grandes amigos, por sua amizade desde a graduação.

Aos professores do IME, que em muito contribuíram para minha formação.

À Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP) pelo apoio financeiro durante todo o período do trabalho, sob o projeto *n*º 2024/01000-9.

Resumo

Marco Antonio Dourado. **Interações não lineares de ondas nas equações de água rasa na esfera**. Dissertação (Mestrado). Instituto de Matemática, Estatística e Ciência da Computação, Universidade de São Paulo, São Paulo, 2025.

Um dos atuais desafios envolvendo a previsão do tempo é o entendimento das interações não lineares entre as dinâmicas rápida e lenta na atmosfera, as quais estão associadas tanto a problemas numéricos, como a escolha do passo de tempo para a estabilidade do método, quanto à modelagem de fenômenos atmosféricos como a Oscilação de Madden Julian. Utilizando o modelo de Água Rasa na esfera, no qual ocorrem tanto ondas rápidas (inerciais-gravidade) quanto lentas (Rossby-Haurwitz) foram analisadas as interações não lineares das ondas em sistemas reduzidos com três, quatro e cinco ondas. Para tanto, os modos normais das Equações de Água Rasa na esfera, dados em termos dos Harmônicos de Hough, foram obtidos. Considerando uma série de Galerking como solução do sistema não linear, equações para a dinâmica de cada modo foram derivadas, além de condições necessárias na estrutura zonal e meridional dos modos para que três ondas interajam. A partir de um sistema de três ondas, transferências de energia entre ondas de Rossby-Haurwitz e gravidade podem ser observadas, sendo que as primeiras são responsáveis pela modulação do período de trocas de energia. Utilizando as soluções analíticas do sistema de três ondas, é possível caracterizar o chamado *pump mode*, além de obter expressões para a eficiência dos modos em um tripleto resonante. Finalmente, em um sistema de quatro ou cinco ondas, o papel de uma onda de gravidade em alterar tanto os campos de energia cinética quanto os períodos de troca em um tripleto de ondas de Rossby-Haurwitz é analisado. Além disso, mostramos a importância das interações não lineares na correta evolução de ondas de Rossby-Haurwitz relevantes atmosféricamente, com relação à alteração de seus campos de energia e períodos de troca de energia.

Palavras-chave: Interações não lineares. Água Rasa. Ondas de gravidade. Ondas de Rossby-Haurwitz.

Abstract

Marco Antonio Dourado. **Nonlinear wave interactions in rotating shallow water equations on the sphere**. Thesis (Master's). Institute of Mathematics, Statistics and Computer Science, University of São Paulo, São Paulo, 2025.

One of the challenges in weather forecasting is the understanding of the nonlinear interactions between the fast and slow dynamics in the atmosphere, which are related both to numerical problems, such as the choice of a stable time-step, along with modeling and understanding the dynamics of atmospheric phenomena, such as the Madden Julian Oscillation. Using a Rotating Shallow Water model on the sphere, in which both fast (inertia-gravity) and slow (Rossby-Haurwitz) waves occur, the nonlinear interactions in reduced models containing three, four and five waves were analyzed. To this end, the normal modes of the Shallow Water Equations on the sphere, given in terms of the Hough harmonics, were obtained. Considering a Galerking expansion as a solution of the nonlinear system, equations for the dynamics of each mode were derived, along with necessary conditions in the zonal and meridional structure of the modes for three interacting waves. From a three-wave system, energy transfer between Rossby-Haurwitz and gravity waves can be observed, where the former modulates the period of energy exchanges. Using the analytical solutions of the three-wave system, the pump mode is characterized, and expressions for the efficiency of the modes in a resonant triad are obtained. Finally, in a four- and five-wave system, the role of a gravity wave in altering the kinetic energy fields and the periods of a Rossby-Haurwitz triad is analyzed. Overall, we show the importance of nonlinear wave interactions in the correct evolution of atmospherically relevant Rossby-Haurwitz waves, with respect to altering their energy fields and periods of energy transfer.

Keywords: Nonlinear interactions. Shallow Water. Gravity waves. Rossby-Haurwitz waves.

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Introduction

As a fluid system, the atmosphere and ocean are described by the *Equations of Motion*, also known as the *Navier-Stokes* equations, which assert the conservation of mass and energy, along with the balance of momentum (cf. CHORIN and MARDSEN, 1993). In particular, their dynamics are defined by the interaction among different wave time-scales, i.e., among fast and slow waves. Examples of fast waves are acoustic and inertia-gravity waves, characterized by high-frequency oscillations, while the Rossby-Haurwitz waves are a type of slow-frequency wave (cf. PEDLOSKY, 1979).

The previously mentioned waves are the eigenfunctions of linearized models describing the atmosphere and ocean, and are also known as *normal modes* of the corresponding system. Nonlinearities break the independence of the modes, enabling the exchange and transfer of energy between waves, generating energy cascades among different scales (cf. M. D. BUSTAMANTE *et al.*, 2014). Furthermore, the interaction between inertia-gravity and Rossby waves is related to the numerical challenge of choosing a time step for the integration of the system. Since the frequency of gravity waves is high, the Courant-Friedrichs-Lewy (*CFL*) condition implies a very small time step for stability, which would result in a higher computational cost. A common solution is to filter out the gravity waves (cf. BONATTI and DIAS, 1983, SALMON, 1988), considering models such as the *quasi-geostrophic* (cf. ZEITLIN, 2018) or the *barotropic non-divergent* equations (cf. HAURWITZ, 1940), whose normal modes are the low-frequency Rossby-Haurwitz waves, although the understanding of the dynamics of such interactions could lead to some alternatives to the time step problem and the consideration of more complex models.

Nonlinear wave interactions among atmospheric waves have been investigated in the literature using different models. In particular, FJORTOFT, 1953 showed that interactions are only possible with three or more waves, since the kinetic energy must change to at least three different scales. These triadic interactions in the *f*-plane, where the Coriolis parameter is assumed to be constant, can be found in the book of MAJDA, 2000, constituting a simple framework where only gravity and geostrophically balanced waves exist. In the context of spherical geometry, RAPHALDINI *et al.*, 2022 studied interactions among four and five waves in the barotropic non-divergent vorticity equations using results about the stability of the modes and coupling coefficients derived by LYNCH, 2009.

However, in the *f*-plane, the slow (Rossby-Haurwitz) waves are degenerate with null frequency, and in the barotropic non-divergent model there are only low-frequency waves, and consequently the exchange of energy among different scales is not precisely considered in these models. In Cartesian coordinates, a first model comprehending fast and

slow dynamics is the *equatorial β -plane*, used by RIPA, 1981, who derived the eigenmodes of the system using *parabolic cylinder functions*, in addition to calculating the coupling coefficients of three waves interacting. RIPA, 1983 described all possible interacting triads in the equatorial β -plane, defining the *pump mode*, i.e., the mode which conceals or receives energy from the other two, as the mode with intermediate slowness. As shown by RAUPP and DIAS, 2006, using the previous formalism, the triadic interaction may explain aspects of intraseasonal oscillations in the tropics, besides shedding light on the phenomena of low-frequency oscillations, known as the *Madden-Julian Oscillation*.

The equatorial β -plane assumes a first-order approximation of the *Coriolis parameter* (f) considering the Earth locally as a plane. Taking into account the sphericity of the planet, the *Rotating Shallow Water* (RSW) equations written in *spherical coordinates* are a first global model which has fast (gravity) waves and slow (Rossby-Haurwitz) waves in its spectrum (cf. VALLIS, 2006). It is worth mentioning that the connection between low and high frequencies is characterized in the RSW equations by the occurrence of the *Kelvin* and *mixed Rossby-gravity* waves, whose existence is related to complicated topological features of the system, as described by DELPLACE *et al.*, 2017.

Since the RSW equations are an approximation of the Navier-Stokes equations, a more complete set of equations modeling the atmosphere and ocean exists, such as the primitive equations and *non-hydrostatic* models (cf. KASAHARA and QUIAN, 2000, RAUPP, TERUYA, *et al.*, 2019). However, the RSW equations constitute a simple prototype for the atmosphere and ocean, and they are an important tool to study the interaction among waves before expanding to a more complex framework. Although VANNEST and VIAL, 1994 have studied the nonlinear interactions between Rossby and inertia-gravity waves in the Shallow Water regime, the approximation to a slowly rotating sphere was considered, and the spectral model consists of the stream function and potential equations, so it is not effectively the general RSW model. Therefore, to our knowledge, there is no literature on triadic interactions in the complete RSW equations on the sphere, and this model could be used, for instance, to study the exchange of energy among Rossby and gravity waves in the *precession resonance regime* (cf. M. D. BUSTAMANTE *et al.*, 2014).

The normal modes of the RSW on the sphere were extensively studied by LONGUET-HIGGINS, 1968, and are the eigenfunctions of the *Laplace Tidal equations*. They can be written in terms of *Hough harmonics*, which do not present a known analytical representation, and therefore are commonly expanded in terms of *spherical vector harmonics*, as considered by SWARZTRAUBER and KASAHARA, 1985, who derived a method, via an infinite system, to calculate the normal modes.

In the present work, the role of inertia-gravity waves in altering the dynamics of Rossby-Haurwitz waves via nonlinear interactions is studied using the RSW model. In particular, relevant modifications in the kinetic energy field and the generation of longer exchanges, whose periods exceed 50 days, are analyzed.

Firstly, in Chapter 1, the RSW model is presented considering its importance in weather forecasting. The dimensionless form of the model is derived along with the conservation of potential vorticity and total energy. In Chapter 2, the Hough harmonics are introduced, expressions for the coupling coefficients and conditions for the nonlinear interactions depending on the symmetry of the modes are obtained. Considering only three modes

interacting nonlinearly, resonant triads and the excitation of Rossby-Haurwitz modes by inertia-gravity waves are analyzed. At the end of the chapter, a characterization of the pump mode is presented using analytical solutions. The four- and five-wave configurations are considered in Chapter 3, where the impacts of a inertia-gravity wave in the energy fields and periods of slow waves are observed. Finally, in Chapter 4 the conclusions and suggestions for future works are presented.

Chapter 1

Governing equations

The *Rotating Shallow Water* (RSW) equations are a particular set of equations describing the dynamics in the atmosphere and ocean by considering them as a thin layer of constant density fluid, in which the horizontal extent is much larger than the layer depth. Once the vertical scale is small, the *hydrostatic approximation* is satisfied, and the mass continuity equation is determined by the fluid column height (cf. PEDLOSKY, 1979).

Assuming a constant density may not well represent phenomena depending on stratification effects, although splitting the atmosphere and ocean into different layers governed by the RSW equations is a first approximation for a stratified fluid. In fact, the *Boussinesq* equations, considering the density field as a small perturbation, mainly in the vertical direction, of a constant value, can be approximated as a *multi-layer* RSW model, with the former set of equations being a simple model where the *buoyancy force* can be studied (cf. HOLTON, 2004).

Furthermore, a two-layer RSW model is a prototype for studying the so-called *baroclinic instability*, which is associated with vertical shear and grows by converting potential energy due to the mean horizontal temperature gradient (cf. HOLTON, 2004). In particular, using the *Quasi-geostrophic* equations, which are an *asymptotic limit* for the RSW model (cf. Madja), necessary conditions for the occurrence of baroclinic instability can be obtained (cf. VALLIS, 2006).

The importance of the RSW model also appears properly in the weather forecast. The role of the gravity waves in the initialization process of weather forecast models, generating instabilities in the finite difference schemes used in the models, and the methods of filtering out these modes, were first studied in the shallow water model, as in the work of MACHENHAUER, 1977. Besides that, the vertical discretization of the *primitive equations* used in numerical implementations leads to a multi-layer RSW model (cf. ZEITLIN, 2018).

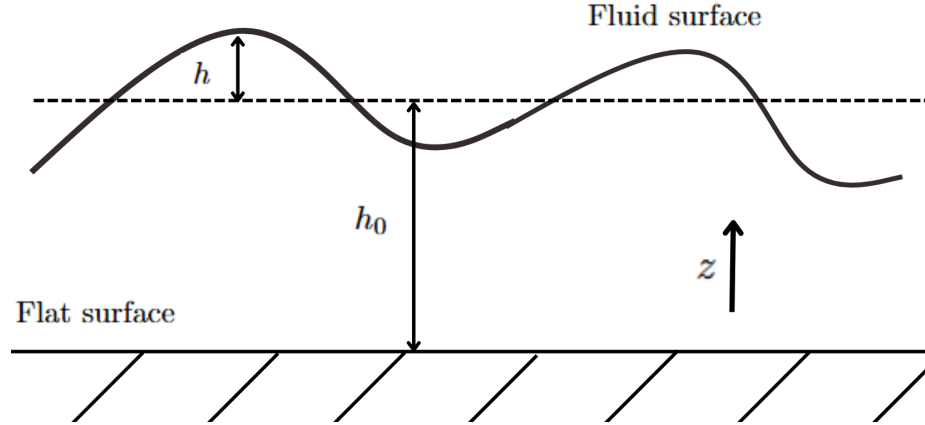


Figure 1.1: A shallow water system. h_0 is the mean height and h the perturbation of the mean height field. It is assumed a flat topography (lower surface).

1.1 Rotating Shallow Water equations

The inviscid one-layer RSW model with a flat lower surface is written in Cartesian coordinates as (cf. PEDLOSKY, 1979)

$$\frac{Du}{Dt} - fv + g \frac{\partial h}{\partial x} = 0, \quad (1.1)$$

$$\frac{Dv}{Dt} + fu + g \frac{\partial h}{\partial y} = 0, \quad (1.2)$$

$$\frac{Dh}{Dt} + (h + h_0) \nabla \cdot \vec{u} = 0, \quad (1.3)$$

where

$$\frac{D\psi}{Dt} = \frac{\partial \psi}{\partial t} + \vec{u} \cdot \nabla \psi,$$

is the *material derivative* for an arbitrary scalar field ψ , $\vec{u} = (u, v)$ is the horizontal velocity, h_0 is the mean height, h the perturbation of the mean height field, f is the Coriolis parameter, and g is the gravitational acceleration.

Considering the Earth's rotation and sphericity, the RSW model written in spherical coordinates treats the globe as a whole, and letting the Coriolis parameter be without approximations (e.g., the f -plane or β -plane approximations), this model yields a better representation of the dynamics and effects of the rotation velocity.

In this context, the RSW equations are given by (cf. [VALLIS, 2006](#))

$$\frac{Du}{Dt} - \left(2\Omega \sin \phi + \frac{u \tan \phi}{a} \right) v + \frac{g}{a \cos \phi} \frac{\partial h}{\partial \lambda} = 0, \quad (1.4)$$

$$\frac{Dv}{Dt} + \left(2\Omega \sin \phi + \frac{u \tan \phi}{a} \right) u + \frac{g}{a} \frac{\partial h}{\partial \phi} = 0, \quad (1.5)$$

$$\frac{Dh}{Dt} + \frac{h + h_0}{a \cos \phi} \left(\frac{\partial u}{\partial \lambda} + \frac{\partial(v \cos \phi)}{\partial \phi} \right) = 0, \quad (1.6)$$

where λ is the longitude, ϕ the latitude, Ω is the Earth's rotation rate, the Coriolis parameter is $f = 2\Omega \sin \phi$, u is the zonal velocity, v the meridional velocity, and the material derivative in spherical coordinates writes

$$\frac{D\psi}{Dt} = \frac{\partial \psi}{\partial t} + \frac{u}{a \cos \phi} \frac{\partial \psi}{\partial \lambda} + \frac{v}{a} \frac{\partial \psi}{\partial \phi}. \quad (1.7)$$

It is convenient to define dimensionless variables (cf. [SWARZTRAUBER and KASAHARA, 1985](#))

$$u' = \frac{u}{\sqrt{gh_0}}, \quad v' = \frac{v}{\sqrt{gh_0}}, \quad h' = \frac{h}{h_0}, \quad t' = 2\Omega t, \quad (1.8)$$

where the mean height (h_0) is considered as a vertical scale, the pure inertia-gravity wave velocity ($\sqrt{gh_0}$) as a velocity scale, and the inverse of the rotation rate (2Ω) as a time scale.

After dropping primes, the system (1.4)-(1.6) is rewritten as

$$\frac{\partial u}{\partial t} + \gamma \left(\frac{u}{\cos \phi} \frac{\partial u}{\partial \lambda} + v \frac{\partial u}{\partial \phi} \right) - (\sin \phi + \gamma u \tan \phi) v + \frac{\gamma}{\cos \phi} \frac{\partial h}{\partial \lambda} = 0, \quad (1.9)$$

$$\frac{\partial v}{\partial t} + \gamma \left(\frac{u}{\cos \phi} \frac{\partial v}{\partial \lambda} + v \frac{\partial v}{\partial \phi} \right) + (\sin \phi + \gamma u \tan \phi) u + \gamma \frac{\partial h}{\partial \phi} = 0, \quad (1.10)$$

$$\frac{\partial h}{\partial t} + \gamma \left(\frac{u}{\cos \phi} \frac{\partial h}{\partial \lambda} + v \frac{\partial h}{\partial \phi} \right) + \frac{(1+h)\gamma}{\cos \phi} \left(\frac{\partial u}{\partial \lambda} + \frac{\partial(v \cos \phi)}{\partial \phi} \right) = 0, \quad (1.11)$$

where

$$\gamma = \frac{\sqrt{gh_0}}{2\Omega a}.$$

The parameter γ is related to the so-called *Lamb's number* (ϵ) by

$$\epsilon = \frac{1}{\gamma^2} = \frac{4a^2\Omega^2}{gh_0} = \left(\frac{a}{\sqrt{gh_0}/(2\Omega)} \right)^2, \quad (1.12)$$

where the Lamb's number measures the influence of the Earth's rotation velocity ($2\Omega a$) and gravity waves ($\sqrt{gh_0}$) on the dynamics.

The linear part of the system (1.9)-(1.11) can be represented by the linear operator

$$\mathcal{L} = \begin{pmatrix} \partial_t & -\sin \phi & \frac{\gamma}{\cos \phi} \partial_\lambda \\ \sin \phi & \partial_t & \gamma \partial_\phi \\ \frac{\gamma}{\cos \phi} \partial_\lambda & \frac{\gamma}{\cos \phi} \partial_\phi ((\cdot) \cos \phi) & \partial_t \end{pmatrix}, \quad (1.13)$$

while the non-linear part by the *bilinear* operator

$$\mathcal{N}(\vec{u}_1, \vec{u}_2) = \gamma \begin{pmatrix} -\frac{u_1}{\cos \phi} \frac{\partial u_2}{\partial \lambda} - v_1 \frac{\partial u_2}{\partial \phi} + u_1 v_2 \tan \phi \\ -\frac{u_1}{\cos \phi} \frac{\partial v_2}{\partial \lambda} - v_1 \frac{\partial v_2}{\partial \phi} - u_1 u_2 \tan \phi \\ -\frac{u_1}{\cos \phi} \frac{\partial h_2}{\partial \lambda} - v_1 \frac{\partial h_2}{\partial \phi} - \frac{h_2}{\cos \phi} \left(\frac{\partial u_1}{\partial \lambda} + \frac{\partial(v_1 \cos \phi)}{\partial \phi} \right) \end{pmatrix}, \quad (1.14)$$

Hence, the RSW equations may be rewritten in an operator form as

$$\mathcal{L}\vec{u} = \mathcal{N}(\vec{u}, \vec{u}). \quad (1.15)$$

1.2 Conserved Quantities

A fluid can conserve properties such as energy, enstrophy and potential vorticity, and there are mainly two types of conservation: material and integral conservation. The former occurs when a property is conserved on each fluid element, in a manner that its material derivative vanishes, while the latter follows an integration over some volume (cf. [VALLIS, 2006](#)).

1.2.1 Potential Vorticity

An example of material conservation in the RSW model is the *potential vorticity* conservation. Considering the *vorticity* of the fluid as the curl of the velocity

$$\zeta = \vec{k} \cdot \nabla \times \vec{u} = \frac{1}{a \cos \phi} \left(\frac{\partial v}{\partial \lambda} - \frac{\partial}{\partial \phi} (u \cos \phi) \right), \quad (1.16)$$

the potential vorticity is defined as

$$Q = \frac{h_0}{h + h_0} (\zeta + f),$$

and its conservation asserts that fluid rotations are in balance with Earth's rotation, enabling the wave-like format observed in the wind field.

Furthermore, the conservation of potential energy is used to derive the so-called *Quasi-geostrophic* (QG) equations on the sphere (cf. [SCHUBERT et al., 2009](#)), a generalization of the *barotropic nondivergent* model. Although further approximations are made to derive the QG equations, yielding just slow (Rossby-Haurwitz) waves in the model, [SCHUBERT et al., 2009](#) and [VERKLEY, 2009](#) showed that for large values of the Lamb's number, and therefore apart from the barotropic nondivergent model, the waves encountered in this

model approximate satisfactorily the waves present in the RSW model.

It is worth mentioning that in the early days of weather forecasting, the barotropic model was used for numerical predictions. However, as the forecasts became hemispheric, this model produced an erroneous westward propagation for ultralong waves when compared with observational data. Heuristic solutions were pointed out by [WOLFF, 1958](#) and [CRESMANN, 1958](#), but a more theoretical alternative was presented by [SCHUBERT *et al.*, 2009](#), who proposed a *correction factor* depending on the QG frequency of the waves, in a manner that the QG frequency, being smaller than the barotropic nondivergent one, would provide a non-expressive westward displacement of ultralong waves.

Finally, in order to derive the potential vorticity conservation, consider the first equation (1.4) rewritten in the vector form as

$$\frac{\partial u}{\partial t} + \vec{u} \cdot \nabla u - \left(f + \frac{u \tan \phi}{a}\right)v + \frac{g}{a \cos \phi} \frac{\partial h}{\partial \lambda} = 0, \quad (1.17)$$

where $\vec{u} = (u, v)$.

Multiplying (1.17) by $\cos \phi$, one has

$$\begin{aligned} & \frac{\partial}{\partial t}(u \cos \phi) + \vec{u} \cdot \nabla(u \cos \phi) + \frac{vu \sin \phi}{a} - \left(f \cos \phi + \frac{u \sin \phi}{a}\right)v + \frac{g}{a} \frac{\partial h}{\partial \lambda} \\ &= \frac{\partial}{\partial t}(u \cos \phi) + \vec{u} \cdot \nabla(u \cos \phi) - fv \cos \phi + \frac{g}{a} \frac{\partial h}{\partial \lambda} = 0, \end{aligned} \quad (1.18)$$

and calculating the partial derivative with respect to the latitude (ϕ), one arrives at

$$\begin{aligned} & \frac{D}{Dt} \left(\frac{\partial}{\partial \phi}(u \cos \phi) \right) + \frac{\partial \vec{u}}{\partial \phi} \cdot \nabla(u \cos \phi) + \frac{\tan \phi}{a} u \frac{\partial u}{\partial \lambda} \\ & - f \frac{\partial}{\partial \phi}(v \cos \phi) - (v \cos \phi) \frac{\partial f}{\partial \phi} + \frac{g}{a} \frac{\partial^2 h}{\partial \phi \partial \lambda} = 0. \end{aligned} \quad (1.19)$$

The second equation (1.6) has the vector form

$$\frac{\partial v}{\partial t} + \vec{u} \cdot \nabla v + \left(f + \frac{u \tan \phi}{a}\right)u + \frac{g}{a} \frac{\partial h}{\partial \phi} = 0, \quad (1.20)$$

and proceeding with the partial derivative with respect to the longitude (λ), one has

$$\frac{D}{Dt} \left(\frac{\partial v}{\partial \lambda} \right) + \frac{\partial \vec{u}}{\partial \lambda} \cdot \nabla v + f \frac{\partial u}{\partial \lambda} + \frac{2 \tan \phi}{a} u \frac{\partial u}{\partial \lambda} + \frac{g}{a} \frac{\partial^2 h}{\partial \lambda \partial \phi} = 0. \quad (1.21)$$

Noting that $(a \cos \phi)^{-1} \cdot ((1.21) - (1.19))$ equals

$$\begin{aligned}
& \frac{1}{a \cos \phi} \frac{D}{Dt} \left(\frac{\partial v}{\partial \lambda} - \frac{\partial}{\partial \phi} (u \cos \phi) \right) + \frac{Df}{Dt} + \frac{f}{a \cos \phi} \left(\frac{\partial u}{\partial \lambda} + \frac{\partial}{\partial \phi} (v \cos \phi) \right) \\
& + \frac{1}{a \cos \phi} \left(\frac{\partial \vec{u}}{\partial \lambda} \cdot \nabla v - \frac{\partial \vec{u}}{\partial \phi} \cdot \nabla (u \cos \phi) \right) + \frac{\tan \phi}{a^2 \cos \phi} u \frac{\partial u}{\partial \lambda} \\
& = \frac{D\zeta}{Dt} - \frac{v \tan \phi}{a^2 \cos \phi} \left(\frac{\partial v}{\partial \lambda} - \frac{\partial}{\partial \phi} (u \cos \phi) \right) - \frac{Df}{Dt} + f \nabla \cdot \vec{u} \\
& + \frac{1}{a \cos \phi} \left(\frac{\partial \vec{u}}{\partial \lambda} \cdot \nabla v - \frac{\partial \vec{u}}{\partial \phi} \cdot \nabla (u \cos \phi) \right) + \frac{\tan \phi}{a^2 \cos \phi} u \frac{\partial u}{\partial \lambda} \\
& = \frac{D}{Dt} (\zeta + f) + f \nabla \cdot \vec{u} - \frac{v \tan \phi}{a^2 \cos \phi} \left(\frac{\partial v}{\partial \lambda} - \frac{\partial}{\partial \phi} (u \cos \phi) \right) \\
& + \frac{1}{a \cos \phi} \left(\frac{\partial \vec{u}}{\partial \lambda} \cdot \nabla v - \frac{\partial \vec{u}}{\partial \phi} \cdot \nabla (u \cos \phi) \right) + \frac{\tan \phi}{a^2 \cos \phi} u \frac{\partial u}{\partial \lambda}, \tag{1.22}
\end{aligned}$$

and that the following equations hold

$$\begin{aligned}
& \frac{v \tan \phi}{a^2 \cos \phi} \frac{\partial}{\partial \phi} (u \cos \phi) - \frac{1}{a \cos \phi} \frac{\partial \vec{u}}{\partial \phi} \cdot \nabla (u \cos \phi) \\
& = \frac{v \tan \phi}{a^2 \cos \phi} \frac{\partial}{\partial \phi} (u \cos \phi) - \frac{1}{a \cos \phi} \left(\frac{1}{a \cos \phi} \frac{\partial u}{\partial \phi} \frac{\partial}{\partial \lambda} (u \cos \phi) + \frac{1}{a} \frac{\partial v}{\partial \phi} \frac{\partial (u \cos \phi)}{\partial \phi} \right) \\
& = -\frac{1}{a^2 \cos^2 \phi} \frac{\partial}{\partial \phi} (v \cos \phi) \frac{\partial}{\partial \phi} (u \cos \phi) - \frac{1}{a^2 \cos \phi} \frac{\partial u}{\partial \phi} \frac{\partial u}{\partial \lambda}, \tag{1.23}
\end{aligned}$$

$$\begin{aligned}
& -\frac{v \tan \phi}{a^2 \cos \phi} \frac{\partial v}{\partial \lambda} + \frac{1}{a \cos \phi} \frac{\partial \vec{u}}{\partial \lambda} \cdot \nabla v \\
& = -\frac{v \tan \phi}{a^2 \cos \phi} \frac{\partial v}{\partial \lambda} + \frac{1}{a \cos \phi} \left(\frac{1}{a \cos \phi} \frac{\partial u}{\partial \lambda} \frac{\partial v}{\partial \lambda} + \frac{1}{a} \frac{\partial v}{\partial \lambda} \frac{\partial v}{\partial \phi} \right) \\
& = \frac{1}{a^2 \cos^2 \phi} \frac{\partial v}{\partial \lambda} \frac{\partial}{\partial \phi} (v \cos \phi) + \frac{1}{a^2 \cos^2 \phi} \frac{\partial u}{\partial \lambda} \frac{\partial v}{\partial \lambda}, \tag{1.24}
\end{aligned}$$

the final terms in (1.22) take the form

$$\begin{aligned}
& -\frac{v \tan \phi}{a^2 \cos \phi} \left(\frac{\partial v}{\partial \lambda} - \frac{\partial}{\partial \phi}(u \cos \phi) \right) + \frac{1}{a \cos \phi} \left(\frac{\partial \vec{u}}{\partial \lambda} \cdot \nabla v - \frac{\partial \vec{u}}{\partial \phi} \cdot \nabla(u \cos \phi) \right) \\
& + \frac{\tan \phi}{a^2 \cos \phi} u \frac{\partial u}{\partial \lambda} \\
& = -\frac{1}{a^2 \cos^2 \phi} \frac{\partial}{\partial \phi}(v \cos \phi) \frac{\partial}{\partial \phi}(u \cos \phi) + \frac{1}{a^2 \cos^2 \phi} \frac{\partial v}{\partial \lambda} \frac{\partial}{\partial \phi}(v \cos \phi) \\
& - \frac{1}{a^2 \cos \phi} \frac{\partial u}{\partial \phi} \frac{\partial u}{\partial \lambda} + \frac{1}{a^2 \cos^2 \phi} \frac{\partial u}{\partial \lambda} \frac{\partial v}{\partial \lambda} + \frac{\tan \phi}{a^2 \cos \phi} u \frac{\partial u}{\partial \lambda} \\
& = \frac{1}{a^2 \cos^2 \phi} \left(\frac{\partial v}{\partial \lambda} - \frac{\partial}{\partial \phi}(u \cos \phi) \right) \frac{\partial}{\partial \phi}(v \cos \phi) \\
& + \frac{1}{a^2 \cos^2 \phi} \frac{\partial u}{\partial \lambda} \frac{\partial v}{\partial \lambda} - \frac{1}{a^2 \cos^2 \phi} \frac{\partial}{\partial \phi}(u \cos \phi) \frac{\partial u}{\partial \lambda} \\
& = \frac{1}{a^2 \cos^2 \phi} \left(\frac{\partial v}{\partial \lambda} - \frac{\partial}{\partial \phi}(u \cos \phi) \right) \left(\frac{\partial u}{\partial \lambda} + \frac{\partial}{\partial \phi}(v \cos \phi) \right) \\
& = \zeta \nabla \cdot \vec{u}.
\end{aligned}$$

Hence, equation (1.22) yields the desired formula

$$\begin{aligned}
& \frac{D}{Dt}(\zeta + f) + (\zeta + f) \nabla \cdot \vec{u} = 0 \\
& \Rightarrow \frac{h_0}{h + h_0} \left(\frac{D}{Dt}(\zeta + f) + (\zeta + f) \nabla \cdot \vec{u} \right) = 0 \\
& \Rightarrow h_0 \left(\frac{D}{Dt}(\zeta + f) \frac{1}{h + h_0} - \frac{\zeta + f}{(h + h_0)^2} \frac{Dh}{Dt} \right) = 0 \\
& \Rightarrow \frac{D}{Dt} \left(\frac{h_0}{h + h_0} (\zeta + f) \right) = 0 \\
& \Rightarrow \frac{D}{Dt} \left(\frac{\zeta + f}{h + h_0} \right) = 0.
\end{aligned}$$

1.2.2 Total Energy

When considering the RSW model as a *Hamiltonian dynamical system*, quantities known as the *Hamiltonian* and the *Casimirs* are proved to be conserved by the dynamics. As considered by [SCHEPHERD, 1990](#), for the RSW model the Hamiltonian is precisely the *total energy*, while examples of Casimirs are the *circulation* and the *mass*.

In Cartesian coordinates, corresponding to the system (1.1) - (1.3), the Hamiltonian is given by

$$\mathcal{H} = \frac{1}{2} \int \int (H |\vec{u}|^2 + gH^2) dx dy, \quad (1.25)$$

where $H = h + h_0$ is the *total height* (cf. [SCHEPHERD, 1990](#)).

Note that the perturbation in the height field (h) has zero average, since h_0 is the height

average itself. In fact,

$$h_0 = \langle H \rangle = \langle h + h_0 \rangle = \langle h \rangle + \langle h_0 \rangle = \langle h \rangle + h_0 \Rightarrow \langle h \rangle = 0.$$

In spherical coordinates, considering the area element $\delta A \approx a^2 \cos \phi$, the Hamiltonian (1.25) is written as

$$\mathcal{H} = \frac{a^2}{2} \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} (H|\vec{u}|^2 + gH^2) \cos \phi \, d\phi d\lambda.$$

Performing the change of variables (1.8) and dropping commas, one has

$$\begin{aligned} \mathcal{H} &= \frac{a^2}{2} \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} ((hh_0 + h_0) \cdot gh_0|\vec{u}|^2 + g(hh_0 + h_0)^2) \cos \phi \, d\phi d\lambda \\ &= \frac{gh_0^2 a^2}{2} \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} ((h+1)|\vec{u}|^2 + (h+1)^2) \cos \phi \, d\phi d\lambda \\ &= \frac{gh_0^2 a^2}{2} \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} (h|\vec{u}|^2 + (u^2 + v^2 + h^2) + 2h + 1) \cos \phi \, d\phi d\lambda. \end{aligned} \quad (1.26)$$

The perturbation's average is precisely

$$\langle h \rangle = \frac{1}{4\pi} \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} h \cos \phi \, d\phi d\lambda,$$

and therefore, this integral is null. Furthermore, the Hamiltonian is defined apart from constants, in a manner that

$$\frac{1}{4\pi} \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} (h|\vec{u}|^2 + (u^2 + v^2 + h^2)) \cos \phi \, d\phi d\lambda, \quad (1.27)$$

is also a conserved quantity.

Equation (1.27) defines the *total energy*, particularly showing that the RSW system is *cubic*, i.e., the total energy is defined by third-order terms. The quadratic part corresponds to the *kinetic energy*

$$E^{(2)} = \frac{1}{4\pi} \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} (u^2 + v^2 + h^2) \cos \phi \, d\phi d\lambda = \frac{1}{2} \|\vec{u}\|^2, \quad (1.28)$$

while the cubic term is the integral

$$E^{(3)} = \frac{1}{4\pi} \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} h|\vec{u}|^2 \cos \phi \, d\phi d\lambda. \quad (1.29)$$

The total energy will be used in the upcoming chapters to study the transfer of energy

among Rossby-Haurwitz and gravity waves, considering both fluxes of energy: from slow waves to fast waves and vice versa. It is worth mentioning that more traditional approaches are available for the derivation of energy conservation, in particular the standard use of kinetic and potential energy densities (cf. MAJDA, 2000). However, as mentioned by SCHEPHERD, 1990, the Hamiltonian description enables the use of methods of *analytic mechanics*, such as the *Noether's theorem*, which relate conservation laws with the system's symmetries.

1.3 Laplace's Tidal equations

As a final comment on the RSW model, it is of great importance to define the *linearized rotating shallow water equations*, since it will be used in the derivation of the normal modes of the system, providing the dispersion relations and expressions for the velocity and height fields of each wave. Assuming that the variables are perturbations of a hydrostatic and steady basic state with mean height h_0 , that is

$$\begin{aligned}\vec{u}'(\lambda, \phi, t) &= \vec{u}_0 + \vec{u}(\lambda, \phi, t), \\ h'(\lambda, \phi, t) &= h_0 + h(\lambda, \phi, t),\end{aligned}$$

where $\vec{u}_0 = (u_0, v_0) = 0$, the linearization of the RSW model writes

$$\frac{\partial u}{\partial t} - 2\Omega \sin \phi v + \frac{g}{a \cos \phi} \frac{\partial h}{\partial \lambda} = 0, \quad (1.30)$$

$$\frac{\partial v}{\partial t} + 2\Omega \sin \phi u + \frac{g}{a} \frac{\partial h}{\partial \phi} = 0, \quad (1.31)$$

$$\frac{\partial h}{\partial t} + \frac{h_0}{a \cos \phi} \left(\frac{\partial u}{\partial \lambda} + \frac{\partial(v \cos \phi)}{\partial \phi} \right) = 0. \quad (1.32)$$

The system (1.30)-(1.32) is known as *Laplace's Tidal equations*.

Proceeding with the change of variables (1.8), and dropping primes, the Laplace tidal equations take the form (cf. SWARZTRAUBER and KASAHARA, 1985)

$$\frac{\partial u}{\partial t} - \sin \phi v + \frac{\gamma}{\cos \phi} \frac{\partial h}{\partial \lambda} = 0, \quad (1.33)$$

$$\frac{\partial v}{\partial t} + \sin \phi u + \gamma \frac{\partial h}{\partial \phi} = 0, \quad (1.34)$$

$$\frac{\partial h}{\partial t} + \frac{\gamma}{\cos \phi} \left(\frac{\partial u}{\partial \lambda} + \frac{\partial(v \cos \phi)}{\partial \phi} \right) = 0. \quad (1.35)$$

Another way to derive the equations (1.33) - (1.35) is by a straightforward linearization of (1.9) - (1.11). For further considerations, the Laplace tidal equations (1.33) - (1.35) can be written in the operator form as

$$\frac{\partial \vec{u}}{\partial t} + L\vec{u} = 0, \quad (1.36)$$

where the linear operator L is given by

$$L = \begin{pmatrix} 0 & -\sin \phi & \frac{\gamma}{\cos \phi} \partial_\lambda \\ \sin \phi & 0 & \gamma \partial_\phi \\ \frac{\gamma}{\cos \phi} \partial_\lambda & \frac{\gamma}{\cos \phi} \partial_\phi (\cos \phi(\cdot)) & 0 \end{pmatrix},$$

in a manner that

$$\mathcal{L} = \frac{\partial}{\partial t} + L,$$

where $\mathcal{L}$ is the operator (1.13).

Chapter 2

Normal Modes

There are two major approaches to solve numerically the equations governing the atmosphere and ocean. On one hand, a grid of discrete points is considered, and the derivatives are approximated by finite difference schemes. Some issues encountered by this method are the representation of the poles and that the model might not conserve energy and enstrophy (cf. KASAHARA, 1977).

On the other hand, the variables can be expanded in a finite series of *orthogonal functions*, which, for simplicity, may constitute eigenfunctions of the linearized equations, also known as *normal modes*, and therefore, are simplified solutions that can be used to study the properties and main features of the system. This method is called *spectral method* and, since solutions are a linear combination of normal modes, the system can be represented as (nonlinear) interactions of such modes.

Normal modes for the f -plane and β -plane approximations are given in terms of the complex exponential, in a way that the solutions are Fourier series (cf. Madja). On the sphere, approximating the velocity field by its non-divergent part and deriving an equation for the potential vorticity from the RSW equations, one obtains the Quasi-geostrophic equations on the sphere, whose normal modes are given in terms of *spheroidal harmonics* (cf. SCHUBERT *et al.*, 2009, VERKLEY, 2009), with *spherical harmonics* being the eigenmodes for a particular case of the previous equations, the barotropic non-divergent vorticity equations (cf. RAPHALDINI *et al.*, 2022).

For the Rotating Shallow Water equations, the eigenfunctions of its linearized model, i.e., the Laplace tidal equations (1.30) - (1.32), are given in terms of *Hough harmonics*, which were extensively studied by LONGUET-HIGGINS, 1968, who, besides providing a method for the calculation of the eigenmodes, considered the asymptotic form of the solutions for different values of the Lamb's number. The normal modes can be divided into two types: eastward and westward inertia-gravity waves (type 1) and Rossby-Haurwitz waves (type 2).

In the next section, the Hough harmonics are built considering them as the eigenfunctions of the Laplace tidal equations, whose eigenvalues constitute the *frequency spectrum* of the linearized RSW equations on the sphere. Furthermore, this spectrum is characterized

by fast and slow frequency modes, whose connection is due to the existence of the so-called *Kelvin* and *Mixed Rossby-Gravity* waves, which in turn are related with the system's topology (cf. DELPLACE *et al.*, 2017).

2.1 Hough harmonics

Following the work of SWARZTRAUBER and KASAHARA, 1985 and noting that the Laplace tidal model (1.36) is linear with respect to time (t), their solutions can be expressed as a linear combination of complex exponentials

$$\vec{u}(\lambda, \phi, t) = H(\lambda, \phi)e^{-i\omega t}, \quad (2.1)$$

where ω is the dimensionless frequency scaled by the Earth's rotation rate 2Ω .

Substitution of (2.1) into (1.36) yields the eigenvalue problem

$$(L - i\omega)H(\lambda, \phi) = 0, \quad (2.2)$$

that is, H is an eigenfunction of the Laplace tidal operator L , and constitutes a *normal mode* for the RSW equations on the sphere.

Furthermore, the operator L is linear in the longitude (λ), in a way that for each *zonal wavenumber* $m \in \mathbb{Z}$, the function $H(\lambda, \phi)$ has the form

$$H(\lambda, \phi) = \Theta(\phi)e^{im\lambda}, \quad (2.3)$$

where $\Theta(\phi)$ defines the *meridional* structure of the function.

Since for a fixed $m \in \mathbb{Z}$ there exist different meridional mode functions, one uses the index n and α to represent the n^{th} meridional mode of type α , where $\alpha \in \{1, 2, 3\}$ describes eastward inert-gravity (EG) waves ($\alpha = 1$), westward inertia-gravity (WG) waves ($\alpha = 2$) and Rossby-Haurwitz (RH) waves ($\alpha = 3$). With this convention, one writes

$$H_{n,\alpha}^m(\lambda, \phi) = \Theta_{n,\alpha}^m(\phi)e^{im\lambda}.$$

$H_{n,\alpha}^m$ is defined as the *Hough harmonic* of zonal wavenumber m , meridional wavenumber n and type α , while $\Theta_{n,\alpha}^m$ is the *Hough function*, which is related to the velocity and height fields by

$$\Theta_{n,\alpha}^m(\phi) = \begin{pmatrix} u_{l,\alpha}^m(\phi) \\ i v_{l,\alpha}^m(\phi) \\ h_{l,\alpha}^m(\phi) \end{pmatrix}. \quad (2.4)$$

In fact, approximating $H(\lambda, \phi)$ by a series of *spherical vector harmonics* with zonal index m , the substitution of (2.3) into (2.2) yields an infinite system, whose eigenvalues define the frequencies (ω), while its eigenvectors define the factors used in the linear combination of spherical vector harmonics, i.e., the eigenvectors of the associated system define the Hough harmonics. Therefore, for a fixed m there are infinitely many eigenvalues, which are divided into three groups, defining the three types of waves α .

From the method proposed by SWARZTRAUBER and KASAHARA, 1985, the normal modes have either a *symmetric structure*, where both zonal velocity (u) and height (h) are symmetric, while the meridional velocity (v) is anti-symmetric around the equatorial plane, or an *anti-symmetric structure* where u and v are anti-symmetric while h is symmetric around the Equator. Considering the zonal wavenumber (m) and the meridional wavenumber (n), the symmetric Rossby-Haurwitz modes are the ones where $m + n$ is *odd*, while the symmetric inertia-gravity modes have $m + n$ *even*.

An important remark is that the Hough harmonic depends strictly on Lamb's number (ϵ), since this parameter defines the linear operator L .

The case with *zero zonal wavenumber* has its own peculiarities: the frequencies of the gravity waves appear as a pair of positive and negative values with the same magnitude, and the frequencies of Rossby-Haurwitz (rotational) waves are all zero (cf. KASAHARA, 1977).

Rossby-Haurwitz waves are associated with low-frequency oscillations, and these modes are absent in the regime for $m = 0$. Therefore, in order to form a *basis of eigenfunctions*, these missing modes should be described, and, in fact, they were given by the zonal geostrophic equation. With this end, KASAHARA, 1977 proposed a split into a zonally averaged part and a nonzonal component for Laplace's tidal equations (1.36). The nonzonal part being solved in terms of Hough harmonics, and the zonally averaged component expanded in terms of Legendre polynomials. The Hough functions constituted eigensolutions for the nonzonal component, but the known eigensolutions for the zonally averaged part were not adaptable as expansion functions. However, in the next year, KASAHARA, 1978 was able to derive a complete set of orthonormal Hough functions for the zonal component, where the normalization was via the Gram-Schmidt process.

The method of SWARZTRAUBER and KASAHARA, 1985 where spherical vector harmonics are considered for the expansion of the Hough harmonics has a special attention for the case $m = 0$, where an effective form of computing the orthonormal basis was considered, without the use of the Gram-Schmidt method, being computationally less expensive.

The method presented by SWARZTRAUBER and KASAHARA, 1985 is used in this work to build the Hough harmonics for the triadic interaction analysis in further sections, and the dispersion relation for the three kinds of waves is plotted in Figure 2.1. The series of spherical vector harmonics was truncated in $N = 30$ terms, while the normalization of the normal modes was done by a Gauss-Legendre quadrature with 60 points.

The dispersion relation (2.1) elucidates why there are three different wave types, since three distinct regions are observed in the plane: a first one in the positive quadrant (eastward inertia-gravity waves), a second one in the unlimited forth quadrant (westward inertia-gravity waves) and the third one as a bounded region, asymptotically converging to zero (Rossby-Haurwitz waves).

In the present work, special attention is devoted to the Kelvin mode (1,1,EG) and the first symmetric Rossby-Haurwitz mode (1,2,RH), since, as presented by TERUYA, RAPHAELINI, RAUPP, *et al.*, 2024, the *Madden Julian Oscillation* (MJO) is dominated by the these modes. The MJO, being first documented by MADDEN and JULIAN, 1972, is the characteristic of both atmospheric circulation and moist convection to be organized in planetary scales and

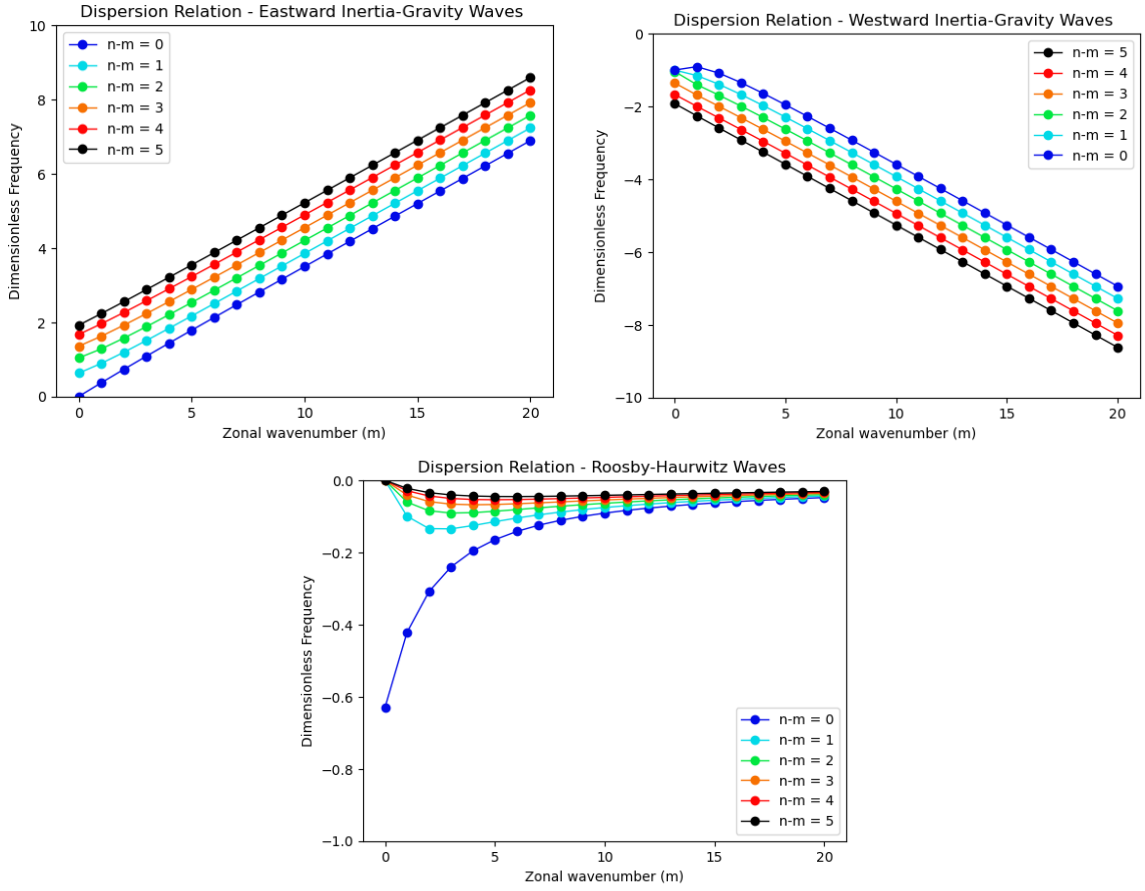


Figure 2.1: Dispersion relation for inertia-gravity and Rossby-Haurwitz waves with a mean height $h_0 = 10\text{km}$. The curves in blue represents the mode $n = m$, and therefore, in the Eastward inertia-gravity waves diagram it represents the Kelvin mode, while in the Rossby-Haurwitz waves diagram, the mixed mode.

to propagate eastward at an average speed of 5m/s with an intraseasonal period ($\sim 30 - 90$ days). In particular, as presented in the extensive review of ZHANG, 2005 on the theme, among other mechanisms, the MJO influences the variability of rainfall over the Pacific islands, modulates the genesis of tropical cyclones and may influence the evolution of El Niño–Southern Oscillation, being clearly characterized by spectral peaks in longitude-time plots of zonal wind and precipitation.

As mentioned by TERUYA, RAPHAELINI, RAUPP, *et al.*, 2024, the spatial structure of the MJO comprehends a symmetric vortical structure about the equatorial plane and a zonal jet structure centered at the Equator. The former is due to the first symmetric Rossby-Haurwitz mode (1,2), while the latter is due to the Kelvin mode (1,1,EG). However, the Kelvin mode has an eastward propagation faster than 5m/s and the Rossby-Haurwitz mode's role is to suppress this velocity, where the group velocity of a multiscale structure would assert the desired speed (cf. ZHANG, 2005).

In Figure 2.2, using the method of SWARZTRAUBER and KASAHARA, 1985, the velocity and height structure of modes (1,1,EG) and (1,2,RH) are presented, which are both *symmetric* around the equatorial plane, and therefore just the positive latitudes (Northern hemisphere)

are shown. Furthermore, since the *barotropic* Rossby-Haurwitz waves are predominant on the MJO, and also are responsible for the tropics–extratropics teleconnections (cf. TERUYA, RAPHALE, RAUPP, *et al.*, 2024), an equivalent height of $h_0 = 10\text{km}$ is chosen for this mode. Even though the *baroclinic* Kelvin mode is the most active in the MJO, the same equivalent height of 10km is chosen, since a two-layer model or the primitive equations would be required to study the interactions among waves with different equivalent heights.

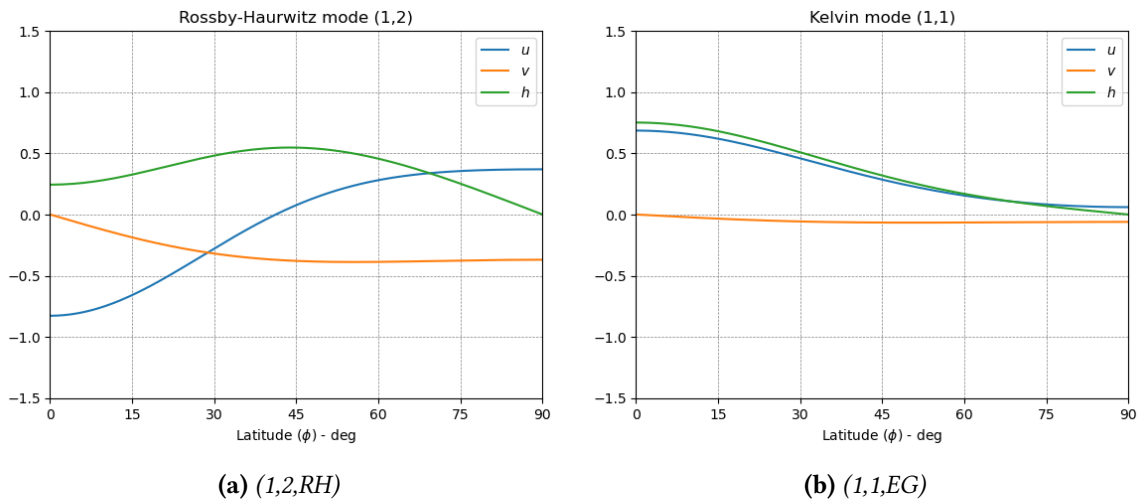


Figure 2.2: Velocity and height structures depending on the latitude (ϕ) for the (a) Rossby-Haurwitz mode (1,2) and (b) Kelvin mode (1,1) for a mean height of $h_0 = 10\text{km}$. Here, the velocity and height fields are in their dimensionless form, following the scales in (1.8).

Another mode that will be considered is the RH wave (4,5). When modeling the atmosphere using the barotropic non-divergent equations, this mode dominates the initial conditions and is used to study the instabilities in the system. In particular, LYNCH, 2009 considering a resonant triad involving mode (4,5,RH) showed that resonant triads enable the breakdown of the aforementioned mode, leading it to an unstable behavior, where the energy exchanges among the waves become chaotic.

The structure of this also symmetric mode is presented in Figure 2.3 for an equivalent height of $h_0 = 10\text{km}$.

2.2 Wave Interactions

Once the normal modes were defined, the solutions of the nonlinear problem (1.15) can be thought of as linear combinations of these modes, in a way that the nonlinearity breaks the independence of the waves, enabling energy exchanges and interactions among them (cf. Madja).

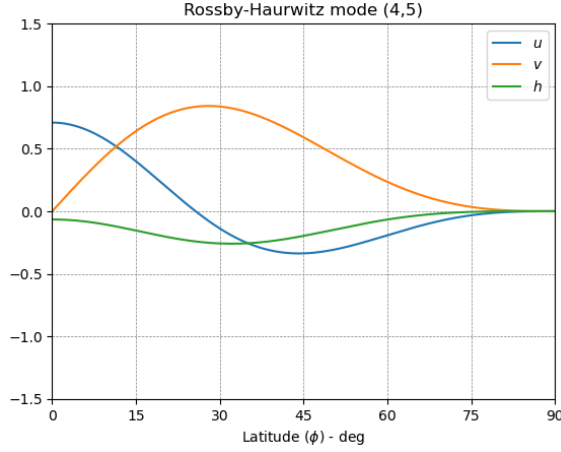


Figure 2.3: Velocity and height structures depending on the latitude (ϕ) for the Rossby-Haurwitz mode (4,5) for a mean height of $h_0 = 10\text{km}$.

Therefore, expanding a solution of (1.15) in terms of Hough harmonics

$$\begin{aligned}\vec{u}(\lambda, \phi, t) &= \sum_{n=0}^{\infty} \sum_{m=0}^n \sum_{\alpha=1}^3 A_{n,\alpha}^m(t) H_{n,\alpha}^m(\lambda, \phi) e^{-i\omega_{n,\alpha}^m t} \\ &= \sum_{n=0}^{\infty} \sum_{m=0}^n \sum_{\alpha=1}^3 A_{n,\alpha}^m(t) \Theta_{n,\alpha}^m(\phi) e^{i(\lambda m - \omega_{n,\alpha}^m t)},\end{aligned}\quad (2.5)$$

the non-linear term (1.14) writes

$$\begin{aligned}\mathcal{N}(\vec{u}, \vec{u}) &= \sum_b A_b(t) e^{-i\omega_b t} \mathcal{N}(\theta_b(\phi) e^{i\lambda m_b}, \vec{u}) \\ &= \sum_b \sum_c A_b(t) A_c(t) e^{-i(\omega_b + \omega_c)t} \mathcal{N}(\Theta_b(\phi) e^{i\lambda m_b}, \Theta_c(\phi) e^{i\lambda m_c}),\end{aligned}\quad (2.6)$$

where a compact notation has been used, in which $b = (m_b, n_b, \alpha_b)$ represents an arbitrary normal mode.

Using the equation (2.2), the bilinear term on the right-hand side of (2.6) can be rewritten as

$$\mathcal{N}(\Theta_b(\phi) e^{i\lambda m_b}, \Theta_c(\phi) e^{i\lambda m_c}) = -i\gamma \begin{pmatrix} \mathcal{B}_1 \\ \mathcal{B}_2 \\ \mathcal{B}_3 \end{pmatrix} e^{i\lambda(m_b + m_c)},$$

where

$$\mathcal{B}_1 = \frac{m_c}{\cos \phi} u_b u_c + v_b \frac{\partial u_c}{\partial \phi} - u_b v_c \tan \phi, \quad (2.7)$$

$$\mathcal{B}_2 = i \frac{m_c}{\cos \phi} u_b v_c + i v_b \frac{\partial v_c}{\partial \phi} - i u_b u_c \tan \phi, \quad (2.8)$$

$$\mathcal{B}_3 = \frac{m_c}{\cos \phi} u_b h_c + v_b \frac{\partial h_c}{\partial \phi} + \frac{m_b}{\cos \phi} u_b h_c + \frac{1}{\cos \phi} h_c \frac{\partial(v_b \cos \phi)}{\partial \phi}. \quad (2.9)$$

Therefore, $\vec{u}(\lambda, \phi, t)$ is a solution of (1.4) - (1.6) provided that

$$\begin{aligned} \mathcal{N}(\vec{u}, \vec{u}) &= \mathcal{L}\vec{u} = \frac{\partial \vec{u}}{\partial t} + L\vec{u} \\ &= \sum_a \frac{dA_a}{dt} H_a e^{-i\omega_a t} - i\omega_a A_a H_a e^{-i\omega t} + \sum_a A_a e^{-i\omega t} L(H_a) \\ &= \sum_a \frac{dA_a}{dt} H_a e^{-i\omega_a t} - i\omega_a A_a H_a e^{-i\omega t} + \sum_a i\omega_a A_a H_a e^{-i\omega t} \\ &= \sum_a \frac{dA_a}{dt} H_a e^{-i\omega_a t}, \end{aligned} \quad (2.10)$$

where the eigenfunction property of Hough harmonics (2.2) was used.

For a single normal mode, the equation (2.10) translates as

$$\begin{aligned} \frac{dA_a}{dt} &= \left\langle \sum_a \frac{dA_a}{dt} H_a e^{-i\omega_a t}, H_a e^{-i\omega_a t} \right\rangle = \langle \mathcal{N}(\vec{u}, \vec{u}), H_a e^{-i\omega_a t} \rangle \\ &= -i\gamma \sum_b \sum_c A_b(t) A_c(t) e^{-i(\omega_b + \omega_c - \omega_a)t} \langle \mathcal{B}, \Theta_a(\phi) e^{im_a \lambda} \rangle, \end{aligned} \quad (2.11)$$

where the orthogonality property of Hough harmonics was used, and the vector $\mathcal{B}$ is defined by

$$\mathcal{B} = (\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3)^t e^{i\lambda(m_b + m_c)},$$

along with the inner product (cf. SWARZTRAUBER and KASAHARA, 1985)

$$\begin{aligned} \langle \Phi, \Psi \rangle &= \frac{1}{2\pi} \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} \Phi \cdot \Psi^* \cos \phi \, d\phi \, d\lambda \\ &= \frac{1}{2\pi} \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} (\Phi_1 \Psi_1^* + \Phi_2 \Psi_2^* + \Phi_3 \Psi_3^*) \cos \phi \, d\phi \, d\lambda. \end{aligned} \quad (2.12)$$

In order for the nonlinear coupling coefficient $\langle \mathcal{B}, \Theta_a e^{im_a \lambda} \rangle$ to be nonzero, the zonal wavenumbers must satisfy

$$m_a = m_b + m_c, \quad (2.13)$$

a constraint due to the longitudinal integral.

Furthermore, the structure on the latitudinal integral cannot be antisymmetric around

the Equator, and therefore, only *three symmetric* modes or *two anti-symmetric* with *one symmetric* mode can interact as is readily seen from (2.7) - (2.9).

2.3 Triadic interaction

A first and simple form to study the energy exchanges among waves, and how waves concede or receive energy from others, is to observe the interactions among *three* modes. As shown by FJORTOFT, 1953, this is the simplest case, where the energy travels through three different scales, constituting a primary example where fast (inertia-gravity) waves and slow (Rossby-Haurwitz) waves can interact.

As an ansatz, consider then just three modes on the expansion (2.5)

$$\begin{aligned}\vec{u}(\lambda, \phi, t) &= A_a(t)\Theta_a(\phi)e^{i(\lambda m_a - \omega_a t)} + A_b(t)\Theta_b(\phi)e^{i(\lambda m_b - \omega_b t)} \\ &+ A_c(t)\Theta_c(\phi)e^{i(\lambda m_c - \omega_c t)} + c.c.,\end{aligned}\quad (2.14)$$

where 'c.c.' indicates the complex conjugate of the previous terms.

It is important to note how the amplitude relates to the initial velocity and height fields. For each mode, the sum of the mode itself with its complex conjugate leads at time $t = 0$ and longitude $\lambda = 0$ to

$$A_a \begin{pmatrix} u_a \\ iv_a \\ h_a \end{pmatrix} + A_a^* \begin{pmatrix} u_a \\ -iv_a \\ h_a \end{pmatrix} = 2 \begin{pmatrix} \text{Re}(A_a)u_a \\ -\text{Im}(A_a)v_a \\ \text{Re}(A_a)h_a \end{pmatrix}, \quad (2.15)$$

and consequently, an initial real amplitude characterizes a zonal jet at the Equator, while a pure imaginary amplitude characterizes a meridional jet. Furthermore, in order to obtain a fixed zonal velocity U m/s, it is sufficient to define

$$A_a = \frac{U}{2\|u_a\| \cdot \sqrt{gh_0}},$$

where the velocity scaling factor $\sqrt{gh_0}$ was used. In particular, $(2\|u_a\| \cdot \sqrt{gh_0})^{-1}$ corresponds to a zonal velocity 1m/s.

Considering that $m_c = m_a + m_b$, equation (2.11) yields the *three-wave equations* (cf. CRAIK, 1988)

$$\frac{dA_a}{dt} = -i\gamma (\langle \mathcal{B}(H_b^*, H_c), H_a \rangle + \langle \mathcal{B}(H_c, H_b^*), H_a \rangle) A_b^* A_c e^{it\delta}, \quad (2.16)$$

$$\frac{dA_b}{dt} = -i\gamma (\langle \mathcal{B}(H_a^*, H_c), H_b \rangle + \langle \mathcal{B}(H_c, H_a^*), H_b \rangle) A_a^* A_c e^{it\delta}, \quad (2.17)$$

$$\frac{dA_c}{dt} = -i\gamma (\langle \mathcal{B}(H_a, H_b), H_c \rangle + \langle \mathcal{B}(H_b, H_a), H_c \rangle) A_a A_b e^{-it\delta}, \quad (2.18)$$

where $\delta = \omega_a + \omega_b - \omega_c$ is the *mismatch* among the waves and the superscript star (**) also indicates the complex conjugate.

Defining new variables

$$A'_a = A_a e^{-it\omega_a}, A'_b = A_b e^{-it\omega_b}, A'_c = A_c e^{-it\omega_c},$$

and the *coupling coefficients*

$$\alpha_{b,c}^a = -\gamma(\langle \mathcal{B}(H_b^*, H_c), H_a \rangle + \langle \mathcal{B}(H_c, H_b^*), H_a \rangle), \quad (2.19)$$

$$\alpha_{a,c}^b = -\gamma(\langle \mathcal{B}(H_a^*, H_c), H_b \rangle + \langle \mathcal{B}(H_c, H_a^*), H_b \rangle), \quad (2.20)$$

$$\alpha_{a,b}^c = -\gamma(\langle \mathcal{B}(H_a, H_b), H_c \rangle + \langle \mathcal{B}(H_b, H_a), H_c \rangle), \quad (2.21)$$

equations (2.16)-(2.18), after dropping primes, can be rewritten in the simplified form

$$\frac{dA_a}{dt} = -i\omega_a A_a + i\alpha_{b,c}^a A_b^* A_c, \quad (2.22)$$

$$\frac{dA_b}{dt} = -i\omega_b A_b + i\alpha_{a,c}^b A_a^* A_c, \quad (2.23)$$

$$\frac{dA_c}{dt} = -i\omega_c A_c + i\alpha_{a,b}^c A_a A_b. \quad (2.24)$$

As mentioned in the previous section, the coupling coefficients are not null provided that the modes a, b, c are all symmetric or constitute a pair of two anti-symmetric modes with one symmetric mode. In particular, this constraint recovers one of the so-called *Ellsaesser conditions*, which provide relations that normal modes in the barotropic non-divergent model (which are only Rossby-Haurwitz modes) must satisfy in order to interact (cf. [RAPHALDINI et al., 2022](#)). The Ellsaesser conditions are:

$$m_c = m_a + m_b, \quad (2.25)$$

$$(m_c)^2 + (m_b)^2 \neq 0, \quad (2.26)$$

$$n_a n_b n_c \neq 0, \quad (2.27)$$

$$n_a + n_b + n_c \text{ is odd}, \quad (2.28)$$

$$(n_a^2 - |m_a|^2) + (n_b^2 - |m_b|^2) > 0, \quad (2.29)$$

$$|n_a - n_b| < n_c < n_a + n_b. \quad (2.30)$$

Since for a symmetric Rossby-Haurwitz wave $m + n$ is odd, three interacting symmetric modes imply by (2.25) that

$$(m_a + n_a) + (m_b + n_b) - (m_c + n_c) \equiv 1 + 1 - 1 \pmod{2}$$

$$n_a + n_b - n_c \equiv 1 \pmod{2}$$

$$n_a + n_b + n_c \equiv 1 + 2n_c \equiv 1 \pmod{2},$$

and the same occurs for two asymmetric with one symmetric mode, recovering the constraint (2.28). It is possible to show that (2.29) and (2.30) are not necessary for a triadic interaction in the RSW model, i.e., there exist three-wave configurations that do not satisfy these constraints, while the coupling coefficients are not null.

Recalling from (1.28) and (1.29) that for the RSW equations on the sphere, the energy

quadratic term is given by

$$E^{(2)} = \frac{1}{2} \langle \vec{u}, \vec{u} \rangle = \frac{1}{4\pi} \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} \vec{u} \cdot \vec{u} \cos \phi \, d\phi \, d\lambda,$$

while the cubic term by

$$E^{(3)} = \frac{1}{4\pi} \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} h(|u|^2 + |v|^2) \cos \phi \, d\phi \, d\lambda,$$

a straightforward calculation yields for the ansatz (2.14)

$$E^{(2)}(t) = |A_a(t)|^2 + |A_b(t)|^2 + |A_c(t)|^2, \quad (2.31)$$

while the cubic term is written as

$$E^{(3)}(t) = 2\text{Re}(A_a^* A_b^* A_c) \int_{-\pi/2}^{\pi/2} [\quad h_a(u_b u_c + v_b v_c) + h_b(u_a u_c + v_a v_c) \\ + \quad h_c(u_a u_b - v_a v_b)] \cos \phi \, d\phi, \quad (2.32)$$

where u_a, v_a, h_a are as in (2.4).

The *conservation of total energy* now is written as

$$\frac{E^{(2)}(t) + E^{(3)}(t)}{E^{(2)}(0) + E^{(3)}(0)} = 1,$$

for all times $t \geq 0$. Noting that

$$\begin{aligned} \frac{d}{dt} |A_a(t)|^2 &= 2\text{Re} \left(A_a^* \frac{d}{dt} A_a(t) \right) \\ &= 2\text{Re} \left(-i\omega_a |A_a|^2 + i\alpha_{b,c}^a A_a^* A_b^* A_c \right) \\ &= 2\alpha_{b,c}^a \text{Im} (A_a A_b A_c^*), \end{aligned} \quad (2.33)$$

and

$$\text{Re} \left(A_b^* A_c \frac{d}{dt} A_a^* \right) = \omega_a \text{Im} (A_a A_b A_c^*),$$

and similarly for b and c , one concludes

$$\frac{d}{dt} E^{(2)}(t) = 2 (\alpha_{b,c}^a + \alpha_{a,c}^b - \alpha_{a,b}^c) \text{Im} (A_a A_b A_c^*),$$

and

$$\frac{d}{dt} E^{(3)}(t) = 2\delta \cdot S_{abc} \cdot \text{Im} (A_a A_b A_c^*), \quad (2.34)$$

where S_{abc} is the integral in (2.32), in a way that the total energy is preserved only if

$$(\alpha_{b,c}^a + \alpha_{a,c}^b - \alpha_{a,b}^c) + \delta S_{abc} = 0. \quad (2.35)$$

2.3.1 Resonant Triads

When the mismatch $\delta = \omega_a + \omega_b - \omega_c$ equals zero in (2.16) - (2.18), the three waves are called a *resonant triad*. As commented on by LYNCH, 2003, the resonance condition ensures that the first two modes in the triad interact non-linearly, producing a total phase corresponding to the third wave, and consequently a *strong interaction* between the three waves is possible. Furthermore, the resonant triads are believed to be associated with the leading-order nonlinear effects (cf. RAPHALDINI *et al.*, 2022), the resonance being possibly one of the main effects generating a higher energy exchange among waves, and consequently the energy cascade.

However, the energy exchanges in a system do not rely only on resonant processes. M. LONGUET-HIGGINS *et al.*, 1967 showed for the β -plane approximation, that when the wave with intermediate wavenumber is a zonal current, the coupling coefficients vanish, and there is no transfer of energy, concluding that the zonal currents cannot gain or lose any energy by the resonance mechanism. Relaxing the resonance condition, REZNIK *et al.*, 1993 considered the role of *quasi-resonant* triads in generating an intensive energy redistribution among Rossby modes. Besides that, considering the stability of the wave field, the nearly resonant triads are also associated with the transition to a turbulent system, where a lower amplitude level is required for these modes to achieve instability (cf. RAPHALDINI *et al.*, 2022). Returning to the RSW model, the same noted by M. LONGUET-HIGGINS *et al.*, 1967 can be observed here: there exist resonant triads that do not interact.

Some examples of quasi-resonant triads of two eastward inertia-gravity modes and a single Rossby-Haurwitz mode whose coupling coefficients vanish are presented in Table 2.1, where an equivalent height of 10km, a typical scale for *barotropic flows* (cf. HOLTON, 2004), is considered. All triads do not satisfy the condition of symmetry in the latitude: the first three triads have two symmetric modes with one anti-symmetric mode, while the fourth has three anti-symmetric modes.

	Triad 1	Triad 2	Triad 3	Triad 4
Mode a	1,6,RH	3,12,RH	3,16,RH	2,10,RH
Mode b	3,7,EG	9,15,EG	10,19,EG	4,11,EG
Mode c	4,7,EG	12,15, EG	13,19, EG	6,11,EG
Mismatch	-9.422e-05	-3.222e-05	-2.642e-05	-6.649e-05

Table 2.1: Examples of quasi-resonant triads, constituted of two eastward inertia-gravity (EG) modes and a single Rossby-Haurwitz (RH) mode, with zero coupling coefficients.

In order to study the dynamics undergone by the kinetic energy fields, noting that the total energy is conserved, the efficiency of transfer is defined as (cf. RAPHALDINI *et al.*, 2022)

$$\mathcal{E} = \frac{\max_t |A_a(t)|^2 - \min_t |A_a(t)|^2}{E_{total}(0)}, \quad (2.36)$$

that is, the efficiency for each mode (here exemplified with mode *a*) is defined as the difference between the maximum and minimum kinetic energy of the desired mode. In particular, the quotient by $E_{total}(0)$ guarantees that the efficiency is bounded by 1, and

therefore it represents the variation in the total energy's percentage centered at each mode.

In the following examples, the system (2.22) - (2.24) is solved using a fourth-order Runge-Kutta method (RK44) with a time step of $h = 0.01$, even though there exist analytic solutions for this system, which will be considered later.

Considering the RH modes (1,2) and (4,5), the resonance condition is almost satisfied when the previous modes are coupled with the RH mode (3,10), the latter being the mode that minimizes the triad mismatch among all the possible triads having the first two RH modes. Furthermore, the RH modes (3,4) and (3,6) appear to be the most excited modes by the former RH modes, i.e., the RH modes (1,2) and (4,5) tend to transfer more energy to these modes, increasing their efficiency.

To illustrate this fact, the two triads presented in Table 2.2 are considered, where the initial energy is projected entirely on the RH modes (1,2) and (4,5), i.e., both RH waves (3,10) and (3,4) are initialized with zero amplitude, and the efficiency of energy transfer in the latter modes is analyzed.

	Triad 1	Freq 1	Coeff. 1
Mode a	3,10,RH	-0.026286	-0.003289
Mode b	1,2,RH	-0.099342	-0.01233
Mode c	4,5,RH	-0.12522	-0.015628
Mismatch		4.85e-04	

	Triad 2	Freq. 2	Coeff. 2
Mode a	3,4,RH	-0.134312	0.443399
Mode b	1,2,RH	-0.099342	-0.028439
Mode c	4,5,RH	-0.12522	0.363881
Mismatch		0.0137	

Table 2.2: Triads with Rossby-Haurwitz modes (1,2) and (4,5).

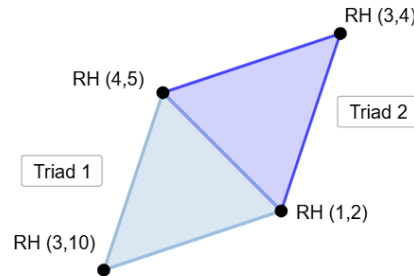


Figure 2.4: Schematic representation of the two triads in Table 2.2.

Considering, for the first triad in Table 2.2, initial amplitudes of the form $(A_a, A_b, A_c) = (0, 0.00232 \times \alpha, 0.00306 \times \beta)$, which, for $\alpha = \beta = 1$ correspond to zonal velocities of 1m/s for the RH modes (1,2) and (4,5), and varying the parameters $\alpha, \beta \in [0, 100]$, it is possible

to study conditions that enhance the excitation of the RH mode (3,10). In particular, Figure 2.5a shows that the maximum efficiency is obtained when the RH mode (1,2) has an initial zonal velocity of approximately 58m/s, while the zonal velocity of the RH mode (4,5) is initially 100m/s, when the efficiency in (3,10) achieves more than 17%. For the initial velocities characterizing the maximum efficiency, Figure 2.5b shows the time integration of the kinetic energy of the three modes, where a period of more than 160 days is obtained, characterizing a long period of interaction, which will be relevant in further discussions about the coupling of triads.

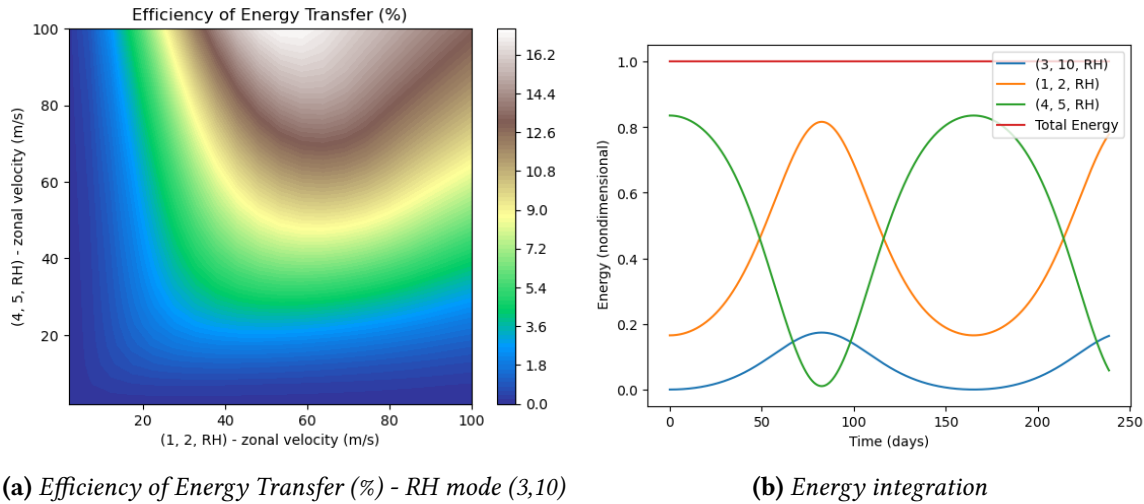


Figure 2.5: Efficiency and integration of triad 1 in Table 2.2 where the Rossby-Haurwitz mode (3,10) is assumed to have a null initial velocity.

On the other hand, when the second triad in Table 2.2 is considered, the energy transfer towards the RH mode (3,4) exceeds 56% in a shorter period of less than 3 days, as observed in Figure 2.6a, where the same range of amplitudes as in the last example were used to build this graph. In Figure 2.6b, one notes that when the RH modes (4,5) and (1,2) have their minimums, a maximum occurs in mode (3,4), characterizing the latter as the *pump mode*, a property that may explain the higher efficiency towards this mode.

The nonlinear interactions, and consequently the energy exchanges, can be stronger (more efficient) for specific wave combinations, i.e., a single mode can interact preferentially with some modes, being almost steady when interacting with others. This fact was considered by the two examples in Table 2.2, where in the presence of the RH mode (3,10), there is a preference for the RH mode (4,5) to interact with the wave (1,2), while when the RH modes (4,5) and (3,4) are coupled, the RH wave (1,2) does not present the same higher efficiency. Therefore, the structure of the waves is determinant for the efficiency of the energy transfers.

Furthermore, the energy can flow through different velocity or length scales, such as the energy transfer from ultra-long waves (Rossby-Haurwitz waves) to waves with a smaller length scale (inertia-gravity waves). However, the inverse flow is also of great interest since it characterizes the energy transfer initially centered on the gravity waves towards the RH waves, and consequently, the routine of filtering out the gravity modes

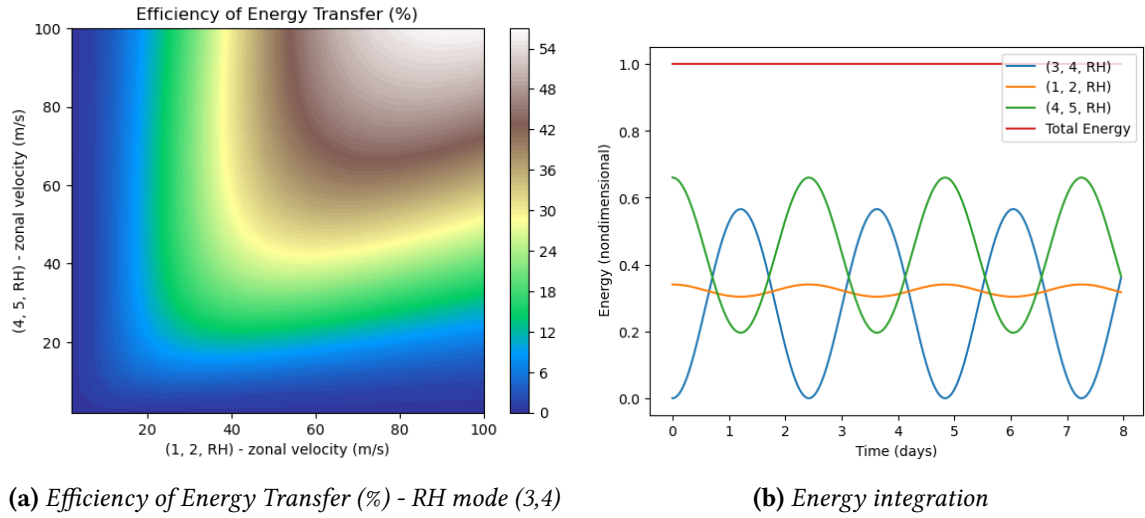


Figure 2.6: Efficiency and integration of triad 2 in Table 2.2 where the Rossby-Haurwitz mode (3,4) is assumed to have a null initial velocity.

in forecast models (cf. [BONATTI and DIAS, 1983](#), [SALMON, 1988](#)) may not be justified if this exchange involves a relevant amount of energy. In fact, the energy due to gravity waves modifies the velocity and height fields of the RH waves, since the kinetic energy can increase (decrease) substantially in these modes and create stronger (weaker) interactions.

The modes presented in Table 2.3 are an example of the energy transfer from RH waves to a gravity wave, where the EG mode (1,1) has an oscillation of more than 9% in its kinetic energy. Even though this efficiency was obtained by considering RH waves with high zonal velocity (see Table 2.3), the efficiency graph in Figure 2.7a shows that for more physically meaningful velocities, relevant efficiencies around 4% can also be obtained, which are represented by the green region around 40m/s. The time integration with the amplitudes for the maximum efficiency is presented in Figure 2.7.

		Coup. coeff.	Freq.	Zonal (m/s)	Amp.
Mode a	1,1,EG	-0.13256	0.3691	80	0.1905
Mode b	3,4,RH	-0.59112	-0.1343	63.89	0.1782
Mode c	4,5,RH	-0.55801	-0.1252	100	0.3061
Mismatch	0.3606				
Efficiency	9.503 %				

Table 2.3: Triad characterizing a flow from RH to EG waves.

Table 2.4 regards the case where the energy flows towards the RH wave. In Figure 2.8 the time integration of the system for the initial amplitudes that maximize the energy transfer to the RH mode is presented. It is important to note that the energy transfer occurs mainly among the gravity modes, where the EG mode (4,4), having initially almost all the system's energy, concedes the same energy to the EG mode (2,4), which can be observed by the peak in the latter mode near 8 days of integration. The RH mode's kinetic energy changes more than 8% during the integration, in a manner that the two EG modes alter

2.3 | TRIADIC INTERACTION

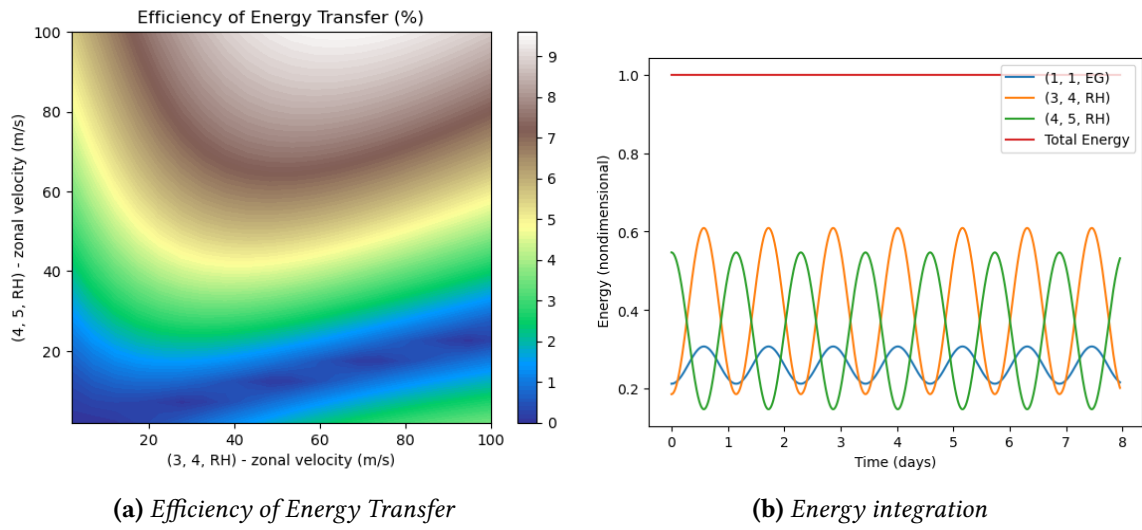


Figure 2.7: Efficiency and time integration of the triad in Table 2.3.

substantially the dynamics of the slow wave.

		Coup. coeff.	Freq.	Zonal (m/s)	Amp.
Mode a	2,3,RH	0.02868	-0.1336	30	0.0756
Mode b	2,4,EG	-0.33704	1.5694	2	0.00649
Mode c	4,4,EG	-0.30817	1.435	100	0.2332
Mismatch	0.00051				
Efficiency	8.3857 %				

Table 2.4: Triad characterizing a flow from EG to RH waves.

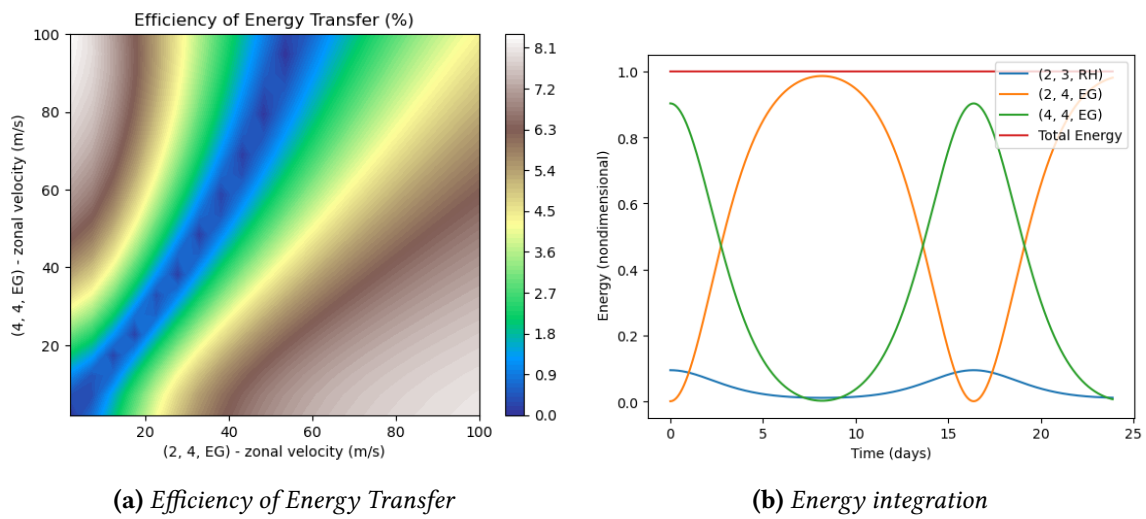


Figure 2.8: Efficiency and time integration of the triad in Table 2.4

2.4 Exact Solutions

Although in the preceding section the integration of the three-wave system was numeric, analytic solutions exist, and their representation sheds light on important features regarding the nonlinear interactions, such as the period of energy exchanges in triads, along with the property of resonant triads in enabling efficiencies close to 1, i.e., in resonant triads the pump mode can receive and concede all the systems' energy.

The exact solutions for the three waves system (2.16) - (2.18), which is written in short notation as

$$\frac{dA_a}{dt} = i\alpha_{b,c}^a A_b^* A_c e^{it\delta} \quad (2.37)$$

$$\frac{dA_b}{dt} = i\alpha_{a,c}^b A_a^* A_c e^{it\delta} \quad (2.38)$$

$$\frac{dA_c}{dt} = i\alpha_{a,b}^c A_a A_b e^{-it\delta} \quad (2.39)$$

are given in terms of the *Jacobi elliptic functions* (cf. CRAIK, 1988) and their periodicity is a fundamental characteristic that can be used to study the energy exchange among coupled triads (cf. M. D. BUSTAMANTE *et al.*, 2014), relating this period with conditions that provide a highly efficient energy transfer among waves, enabling, for example, the excitation of Rossby modes by gravity waves and the consequent cascade of energy to small scales. It is worth mentioning that the solutions can also be obtained by defining the so-called *Lax pairs*, as considered in HARRIS, 2013, where the details and a historical review of the matter can be found.

The analytic solutions for the amplitudes also characterize the *pump mode*, i.e., the mode which either receives or transmits energy to the other two modes. As will be clear later in the chapter, the kinetic energy in each mode has the same period, being modulated by the *elliptic sine*, and when the pump mode has a maximum, the others have a minimum in this energy field, which is the main property of the pump mode.

In order to derive the existence of the pump mode, four different cases arise when the coupling coefficients' sign is considered. Assuming, without loss of generality, that $\alpha_{b,c}^a \leq \alpha_{a,c}^b$ the possible sign combinations are separated into different cases as in Table 2.5. In particular, the sign's tuples in Case IV constitute systems without pump modes, which do not occur when the modes are in resonance, since the total energy conservation (2.35) would imply

$$\alpha_{b,c}^a + \alpha_{a,c}^b - \alpha_{a,b}^c = 0, \quad (2.40)$$

and therefore, the coupling coefficient of mode c must have the same sign as at least one of the remaining coupling coefficients.

The following change of variables will be considered in all cases, leading to different

$\alpha_{b,c}^a$		+	-
$\alpha_{a,c}^b$		+	-
$\alpha_{a,b}^c$		+	-

(a) Case I

$\alpha_{b,c}^a$		-
$\alpha_{a,c}^b$		+
$\alpha_{a,b}^c$		-

(b) Case II

$\alpha_{b,c}^a$		-
$\alpha_{a,c}^b$		+
$\alpha_{a,b}^c$		+

(c) Case III

$\alpha_{b,c}^a$		+	-
$\alpha_{a,c}^b$		+	-
$\alpha_{a,b}^c$		-	+

(d) Case IV

Table 2.5: Sign combinations corresponding to Cases I, II, III and IV. For instance, the first column in (a) comprehends the case where all the coupling coefficients are positive.

systems that will be analyzed separately:

$$A_a' = -i\sqrt{\alpha_{a,c}^b \cdot \alpha_{a,b}^c} A_a, \quad (2.41)$$

$$A_b' = -i\sqrt{\alpha_{a,b}^c \cdot \alpha_{b,c}^a} A_b, \quad (2.42)$$

$$A_c' = -i\sqrt{\alpha_{a,b}^a \cdot \alpha_{a,c}^b} A_c. \quad (2.43)$$

2.4.1 Case I

For the sign combinations as in Table 2.5a, the system (2.37) - (2.39) takes the simplified form

$$\frac{dA_a'}{dt} = (A_b')^* A_c' e^{it\delta}, \quad (2.44)$$

$$\frac{dA_b'}{dt} = (A_a')^* A_c' e^{it\delta}, \quad (2.45)$$

$$\frac{dA_c'}{dt} = -A_a' A_b' e^{-it\delta}. \quad (2.46)$$

In the resonant case ($\delta = 0$), the above system is precisely the dynamical system for a complex triad considered by KARTASHOVA and M. BUSTAMANTE, 2011.

Furthermore, in the resonant case, there are two *Manley-Rowe invariants*

$$I_{a,c} = |A_a'|^2 + |A_c'|^2, \quad I_{b,c} = |A_b'|^2 + |A_c'|^2,$$

which are given by

$$I_{a,c} = |A_a|^2 |\alpha_{a,c}^b \cdot \alpha_{a,b}^c| + |A_c|^2 |\alpha_{b,c}^a \cdot \alpha_{a,c}^b|, \quad (2.47)$$

$$I_{b,c} = |A_b|^2 |\alpha_{a,b}^c \cdot \alpha_{b,c}^a| + |A_c|^2 |\alpha_{b,c}^a \cdot \alpha_{a,c}^b|, \quad (2.48)$$

concluding, after a division by $\alpha_{a,b}^c \cdot \alpha_{b,c}^a \cdot \alpha_{a,c}^b$, that

$$I'_{a,c} = \frac{|A_a|^2}{\alpha_{b,c}^a} + \frac{|A_c|^2}{\alpha_{a,b}^c}, \quad I'_{b,c} = \frac{|A_b|^2}{\alpha_{a,c}^b} + \frac{|A_c|^2}{\alpha_{a,b}^c},$$

are also conserved (cf. [RAPHALDINI et al., 2022](#)).

Following [KARTASHOVA and M. BUSTAMANTE, 2011](#) and writing the amplitudes in their *polar representation*

$$A'_j = B'_j e^{i\varphi_j}, \quad (2.49)$$

with $B_j, \varphi_j \in \mathbb{R}$, equation (2.44) yields

$$\begin{aligned} \frac{dB'_a}{dt} e^{i\varphi_a} + i \frac{d\varphi_a}{dt} B'_a e^{i\varphi_a} &= B'_b B'_c e^{i(\varphi_c - \varphi_b + \delta t)} \\ \Rightarrow \frac{dB'_a}{dt} + i \frac{d\varphi_a}{dt} B'_a &= B'_b B'_c e^{i(\varphi_c - \varphi_a - \varphi_b + \delta t)}. \end{aligned} \quad (2.50)$$

The real part of (2.50) gives

$$\frac{dB'_a}{dt} = B'_b B'_c \cos(\Phi_{a,b}^c),$$

while the imaginary part

$$\frac{d\varphi_a}{dt} = \frac{B'_b B'_c}{B'_a} \sin(\Phi_{a,b}^c),$$

where $\Phi_{a,b}^c = \varphi_c - \varphi_a - \varphi_b + \delta t$ is the so-called *dynamical phase* (cf. [RAPHALDINI et al., 2022](#)).

Hence, proceeding similarly for equations (2.45) and (2.46), the three-wave system is rewritten as a four-equation system as

$$\frac{dB'_a}{dt} = B'_b B'_c \cos(\Phi_{a,b}^c), \quad (2.51)$$

$$\frac{dB'_b}{dt} = B'_a B'_c \cos(\Phi_{a,b}^c), \quad (2.52)$$

$$\frac{dB'_c}{dt} = -B'_b B'_c \cos(\Phi_{a,b}^c), \quad (2.53)$$

$$\frac{d\Phi_{a,b}^c}{dt} = -\delta + \mathcal{H} \left(\frac{1}{B_a'^2} + \frac{1}{B_b'^2} - \frac{1}{B_c'^2} \right), \quad (2.54)$$

where the *Hamiltonian* is

$$\mathcal{H} = B'_a B'_b B'_c \sin(\Phi_{a,b}^c). \quad (2.55)$$

Defining new variables

$$\rho = \frac{I_{b,c}}{I_{a,c}},$$

and $\nu \in [0, \pi]$ such that

$$\cos \nu = \frac{(-2 + 3\rho + 3\rho^2 - 2\rho^3)I_{a,c}^3 - 27\mathcal{H}}{2(1 - \rho + \rho^2)^{3/2}I_{a,c}^3},$$

the solution for the moduli B'_j , in the resonant case, can be expressed as (cf. [KARTASHOVA and M. BUSTAMANTE, 2011](#))

$$\begin{aligned} |B'_a(t)|^2 &= -\mu \left(\frac{2K(\mu)}{T} \right)^2 \operatorname{sn}^2 \left(2K(\mu) \frac{t - t_0}{T}, \mu \right) \\ &+ \frac{I_{a,c}}{3} (2 - \rho + 2\sqrt{1 - \rho + \rho^2} \cos(\nu/3)), \end{aligned} \quad (2.56)$$

$$\begin{aligned} |B'_b(t)|^2 &= -\mu \left(\frac{2K(\mu)}{T} \right)^2 \operatorname{sn}^2 \left(2K(\mu) \frac{t - t_0}{T}, \mu \right) \\ &+ \frac{I_{a,c}}{3} (2\rho - 1 + 2\sqrt{1 - \rho + \rho^2} \cos(\nu/3)), \end{aligned} \quad (2.57)$$

$$\begin{aligned} |B'_c(t)|^2 &= \mu \left(\frac{2K(\mu)}{T} \right)^2 \operatorname{sn}^2 \left(2K(\mu) \frac{t - t_0}{T}, \mu \right) \\ &+ \frac{I_{a,c}}{3} (\rho + 1 + 2\sqrt{1 - \rho + \rho^2} \cos(\nu/3)), \end{aligned} \quad (2.58)$$

where $\operatorname{sn}(\cdot, \mu)$ is the *elliptic sine* and $K(\mu)$ is the *elliptic integral of the first kind* (cf. [CRAIK, 1988](#))

$$K(\mu) = \int_0^{\pi/2} \frac{1}{\sqrt{1 - \mu \sin^2 \theta}} d\theta,$$

where

$$\mu = \frac{\cos(\nu/3 + \pi/6)}{\cos(\nu/3 - \pi/6)}.$$

Furthermore,

$$T = \frac{3^{1/4} \sqrt{2} K(\mu)}{(1 - \rho + \rho^2)^{1/4} \sqrt{\cos(\nu/3 - \pi/6)} \sqrt{I_{a,c}}}, \quad (2.59)$$

is the *period* of the amplitudes and dynamical phase, which in contexts, e.g., barotropic non-divergent model or the Charney-Hasegawa-Mima (CHM) equation, is associated with the so-called *precession resonance mechanism*, which asserts conditions involving both triad's period and dynamical phase to generate stronger interactions between a resonant triad coupled with non-resonant triads (cf. [M. D. BUSTAMANTE et al., 2014](#), [RAPHAELINI et al., 2022](#)).

For the non-resonant case, [HARRIS, 2013](#) provides an expression for the analytic solutions, where different conservation laws are defined, instead of the Manley-Rowe invariants, even though a similar formula for the solutions' period is obtained. As the expressions regarding both analytical solution and period of non-resonant triads are well explained and are readily understood from the work of [HARRIS, 2013](#), they are here omitted.

Besides that, the period formula proposed by [HARRIS, 2013](#) (hereafter denoted by T_{Harris}) is more general, since it is not restricted to the resonant case, and some differences are indeed found between his period and the one considered by [KARTASHOVA and M. BUSTAMANTE, 2011](#) (T_{K-H}) when non-resonant triads are considered. For the quasi-resonant triad in Table 2.2, the following periods are obtained:

$$T_{K-H} = 166.302 \text{ days,}$$

$$T_{Harris} = 166.258 \text{ days,}$$

which are satisfactorily close to each other. However, when Triad 2 in Table 2.2 (whose mismatch is $\delta = 0.0137$) has all modes with initial zonal velocities at $25m/s$, the periods are

$$T_{K-H} = 5.2611 \text{ days,}$$

$$T_{Harris} = 3.6404 \text{ days,}$$

with a difference of more than 1 day, which reveals some limitations in the method proposed by [KARTASHOVA and M. BUSTAMANTE, 2011](#) when extended to non-resonant triads. Figure 2.9 shows the time integration for the three modes initialized with a zonal velocity of $25m/s$.

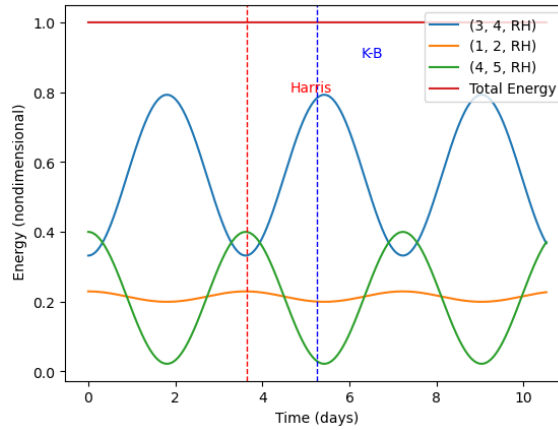


Figure 2.9: Triad 2 in Table 2.2 where all the modes have an initial zonal velocity of $25m/s$.

Undoing the prosecuted change of variables (2.41) - (2.43), and recalling that the kinetic energy is the squared absolute value of the amplitude, one finally obtains the kinetic energy for each mode as

$$|B_a|^2 = \frac{1}{|\alpha_{a,b}^c \cdot \alpha_{a,c}^b|} |B'_a|^2, \quad |B_b|^2 = \frac{1}{|\alpha_{a,b}^c \cdot \alpha_{b,c}^a|} |B'_b|^2, \quad |B_c|^2 = \frac{1}{|\alpha_{b,c}^a \cdot \alpha_{a,c}^b|} |B'_c|^2. \quad (2.60)$$

Furthermore, as equations (2.56) - (2.58) describe, all the kinetic energies are translations of the same function

$$\pm \mu \left(\frac{2K(\mu)}{T} \right)^2 \text{sn}^2 \left(2K(\mu) \frac{t - t_0}{T}, \mu \right), \quad (2.61)$$

and consequently, due to the positive sign in equation (2.58), where the squared elliptic sine has its minimum, corresponding to a maximum for $|A'_a(t)|^2 = |B'_a(t)|^2$ and $|A'_b(t)|^2$, the third wave amplitude moduli has its minimum, characterizing *mode c* as the *pump mode*. In fact, the amplitudes of the original three-wave system are just positive multiples of the amplitudes A'_a , A'_b and A'_c as readily seen for (2.60), in a manner that maxima and minima in $|A_a|^2$, $|A_b|^2$, $|A_c|^2$ are exactly the same for $|A'_a|^2$, $|A'_b|^2$, $|A'_c|^2$.

In the non-resonant regime, the amplitude moduli for all modes are also translations of a single function, labeled by HARRIS, 2013 as $v(t)$. Precisely equal to the resonant case, modes *a* and *b* are governed by the *positive* sign of $v(t)$ where mode *c* by the negative sign ($-v(t)$) which again characterizes mode *c* as the pump mode.

2.4.2 Case II

The change of variables (2.41) - (2.43) leads now to the system

$$\frac{dA'_a}{dt} = (A'_b)^* A'_c e^{it\delta}, \quad (2.62)$$

$$\frac{dA'_b}{dt} = -(A'_a)^* A'_c e^{it\delta}, \quad (2.63)$$

$$\frac{dA'_c}{dt} = -A'_a A'_b e^{-it\delta}, \quad (2.64)$$

and replacing A'_a by $(-A'_a)^*$ and A'_c by its complex conjugate, the above system becomes

$$\frac{d(-A'_a)^*}{dt} = -(A'_b)(A'_c)^* e^{-it\delta}, \quad (2.65)$$

$$\frac{dA'_b}{dt} = (-A'_a)^* A'_c e^{it\delta}, \quad (2.66)$$

$$\frac{d(A'_c)^*}{dt} = (-A'_a)^* (A'_b)^* e^{it\delta}, \quad (2.67)$$

which is precisely the system (2.44) - (2.46) for the variables A'_b , $(A'_c)^*$, $(-A'_a)^*$, i.e., one arrives at Case I.

Since

$$\begin{aligned} |(-A'_a)^*(t)|^2 &= |A'_a(t)|^2, \\ |(A'_c)^*(t)|^2 &= |A'_c(t)|^2, \end{aligned}$$

the system (2.62) - (2.64) implies that mode *a* is described by the negative sign of elliptic sine (2.61) or of $v(t)$, and consequently it is the pump mode for the sign combinations of Case II.

2.4.3 Case III

The same change of variables can be used to derive a system similar to (2.44) - (2.46), however, another argument is here presented to justify the election of mode *b* as the pump mode.

Recalling that the kinetic energy for each mode is given by (see (2.33))

$$\begin{aligned}\frac{d}{dt}|A_a(t)|^2 &= 2\alpha_{b,c}^a \text{Im}(A_a A_b A_c^*), \\ \frac{d}{dt}|A_b(t)|^2 &= 2\alpha_{a,c}^b \text{Im}(A_a A_b A_c^*), \\ \frac{d}{dt}|A_c(t)|^2 &= -2\alpha_{a,b}^c \text{Im}(A_a A_b A_c^*),\end{aligned}$$

for the sign configurations in Case III, one has that

$$\begin{aligned}\frac{d}{dt}|A_a(t)|^2 \cdot \frac{d}{dt}|A_c(t)|^2 &\geq 0, \\ \frac{d}{dt}|A_a(t)|^2 \cdot \frac{d}{dt}|A_b(t)|^2 &\leq 0, \quad \frac{d}{dt}|A_b(t)|^2 \cdot \frac{d}{dt}|A_c(t)|^2 \leq 0.\end{aligned}$$

Therefore, when modes a and c increase, mode b has a negative derivative, implying that critical points (which are the same for all modes - see (2.56) - (2.58)) corresponding to a maximum for a and c are a minimum for mode b , characterizing the latter as the pump mode.

2.4.4 Case IV

The sign combinations reserved for the fourth case do not lead to a system similar to (2.44) - (2.46), and the same analysis used in the third case when made in this case, highlights the fact that all three kinetic energies increase or decrease together, i.e., the derivatives' sign is the same for all three modes. In particular, there is no pump mode.

As an example, consider the triad in Table 2.6 when a nonphysical initial amplitude is chosen just to generate a better visualization of the nonlinear interactions. Figure 2.10 shows explicitly that the kinetic energy's modulation for all three modes is the same, having valleys and peaks at the same time. Furthermore, the kinetic energy, being the sum of the three waves' amplitudes, exceeds the total energy in the peaks, generating a negative cubic energy, which is not impossible, since the cubic energy depends on the height perturbation (h), which can be negative.

		Coup. coeff.	Frequency	Amplitude
Mode a	1,2,RH	0.0103	-0.0993	2
Mode b	2,2,EG	0.7303	0.0906	2
Mode c	3,5,EG	-0.03936	1.8794	2

Table 2.6: Triad whose coupling coefficients are in Case IV.

Finally, all the cases can be synthesized as in Table 2.7, and as a straightforward consequence of equation (2.40), at least for the resonant triads the pump mode is the mode with the *highest coupling coefficient in absolute value*.

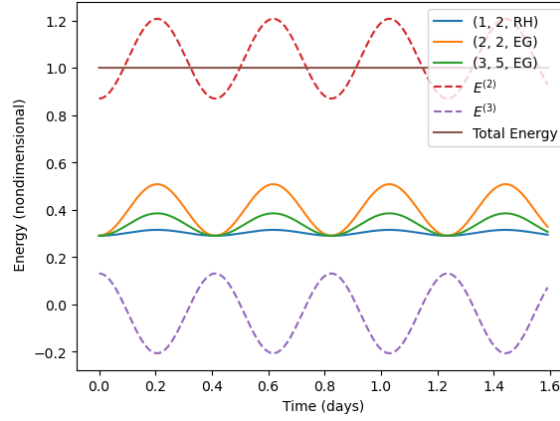


Figure 2.10: Time integration for the triad in Table 2.6. $E^{(2)}$ and $E^{(3)}$ are the quadratic (equation (2.31)) and cubic (equation (2.32)) respectively.

	Case I		Case II	Case III	Case IV	
$\alpha_{b,c}^a$	+	-	-	-	+	-
$\alpha_{a,c}^b$	+	-	+	+	+	-
$\alpha_{a,b}^c$	+	-	-	+	-	+
Pump mode	mode c		mode a	mode b	none	

Table 2.7: Pump modes for the three-wave system depending on the coupling coefficients of the original triad.

2.4.5 A special case: $\mathcal{H} = 0$

Depending on the sign of the coupling coefficients, it is possible to ensure real amplitudes in (2.41) - (2.43) in a manner that the Hamiltonian (2.55) is null, once each individual phase is zero and consequently the dynamical phase also vanishes. It is worth mentioning that this case is always possible, since the initial amplitudes A_a, A_b, A_c in (2.14) can be chosen as any complex number (cf. [RAPHALDINI et al., 2022](#)).

Furthermore, by (2.60), the efficiency of energy transfer for the system (2.37) - (2.39) is, at least for a scaling factor, the same for the system (2.51) - (2.54), and therefore, the analytical solutions (2.56) - (2.58) can be used to study the exchange of energy.

Considering that $t \geq 0$ and also $|\text{sn}(\cdot, \mu)| \leq 1$, the problem of maximizing the energy transfer reduces to maximizing

$$\begin{aligned}
 \max_{t \geq 0} |B'_a(t)|^2 - \min_{t \geq 0} |B'_a(t)|^2 &= \mu \left(\frac{2K(\mu)}{T} \right)^2 \\
 &= \frac{\cos(v/3 + \pi/6)}{\cos(v/3 - \pi/6)} \left(\frac{\sqrt{2}}{3^{1/4}} (1 - \rho + \rho^2)^{1/4} \sqrt{\cos(v/3 - \pi/6)} \sqrt{I_{a,c}} \right)^2 \\
 &= \frac{2}{\sqrt{3}} (1 - \rho + \rho^2)^{1/2} \cos(v/3 + \pi/6) I_{a,c}.
 \end{aligned} \tag{2.68}$$

Using the trigonometric identity

$$\cos(v) = 4 \cos^3\left(\frac{v}{3}\right) - 3 \cos\left(\frac{v}{3}\right), \quad (2.69)$$

and writing $\cos(v/3) = P(\rho)/Q(\rho)$ where $P, Q^2 \in \mathbb{R}[\rho]$ are polynomials, one has

$$\cos v = \frac{(-2 + 3\rho + 3\rho^2 - 2\rho^3)}{2(1 - \rho + \rho^2)^{3/2}} = \frac{4P^3(\rho) - 3P(\rho)Q^2(\rho)}{Q^3(\rho)}, \quad (2.70)$$

which yields

$$Q(\rho) = 2^{1/3}(1 - \rho + \rho^2)^{1/2},$$

and therefore, considering $P(\rho) = a\rho + b$ the numerator in (2.70) provides

$$\begin{aligned} & 4(a^3\rho^3 + 3a^2b\rho^2 + 3ab^2\rho + b^3) - 3 \cdot 4^{1/3}(a\rho^3 + \rho^2(b - a) + \rho(a - b) + b) \\ &= -2 + 3\rho + 3\rho^2 - 2\rho^3, \end{aligned}$$

whose solution is simply given by

$$a = b = 4^{-1/3},$$

in a manner that

$$\cos\left(\frac{v}{3}\right) = \frac{\rho + 1}{2(1 - \rho + \rho^2)^{1/2}},$$

and consequently

$$\sin\left(\frac{v}{3}\right) = \frac{\sqrt{3}}{2} \frac{\rho - 1}{(1 - \rho + \rho^2)^{1/2}},$$

where it is assumed, without loss of generality, that $\rho \geq 1$, i.e., $|A'_a(0)| \leq |A'_b(0)|$; if $\rho \leq 1$, it is sufficient to change the lines of modes a and b in the system, because their roles are the same.

Therefore, the cosine term in the efficiency equation (2.68) now is written as

$$\begin{aligned} \cos(v/3 + \pi/6) &= \frac{\sqrt{3}}{2} \cdot \frac{\rho + 1}{2(1 - \rho + \rho^2)^{1/2}} - \frac{1}{2} \cdot \frac{\sqrt{3}}{2} \frac{\rho - 1}{(1 - \rho + \rho^2)^{1/2}} \\ &= \frac{\sqrt{3}}{2} \frac{1}{(1 - \rho + \rho^2)^{1/2}}, \end{aligned}$$

which finally yields the simple form for the efficiency

$$\max_{t \geq 0} |B'_a(t)|^2 - \min_{t \geq 0} |B'_a(t)|^2 = I_{a,c}.$$

Note that, for the system (2.44) - (2.46) all modes have the same efficiency, in this case given by $I_{a,c}$, although for the original system, this efficiency is modulated by the factors depending on the coupling coefficients as in (2.41) - (2.43), which also asserts the total energy conservation. Therefore, for a resonant triad it is possible to achieve an efficiency near to 100% by choosing higher initial amplitudes, which, however, can lead to

nonphysical solutions.

The efficiency (2.36) in each mode is given by

$$\mathcal{E}_a = \frac{1}{E_{total}(0)} \left(\frac{I_{a,c}}{|\alpha_{a,c}^b \alpha_{a,b}^c|} \right) = \frac{1}{E_{total}(0)} \left(|A_a|^2 + \frac{|\alpha_{b,c}^a|}{|\alpha_{a,b}^c|} |A_c|^2 \right), \quad (2.71)$$

$$\mathcal{E}_b = \frac{1}{E_{total}(0)} \left(\frac{|\alpha_{a,c}^b|}{|\alpha_{b,c}^a|} |A_a|^2 + \frac{|\alpha_{a,c}^b|}{|\alpha_{a,b}^c|} |A_c|^2 \right), \quad (2.72)$$

$$\mathcal{E}_c = \frac{1}{E_{total}(0)} \left(\frac{|\alpha_{a,b}^c|}{|\alpha_{b,c}^a|} |A_a|^2 + |A_c|^2 \right), \quad (2.73)$$

where (2.47) were used.

As an example, consider the quasi-resonant triad constituted of a single RH mode coupled with two EG modes as in Table 2.8. The aforementioned condition $|A'_a(0)| < |A'_b(0)|$ is translated as

$$|\alpha_{a,c}^b \alpha_{a,b}^c| |A_a(0)|^2 < |\alpha_{b,c}^a \alpha_{a,b}^c| |A_b(0)|^2 \Rightarrow \left| \frac{A_a(0)}{\alpha_{b,c}^a} \right| < \left| \frac{A_b(0)}{\alpha_{a,c}^b} \right|,$$

and since $|\alpha_{b,c}^a| \gg |\alpha_{a,c}^b|$, this condition is satisfied for reasonable values of initial amplitudes, which justifies the choice of the EG wave (6,9) as the mode a, and not the wave with the lowest zonal wavenumber.

		Coup. coeff.	Frequency
Mode a	6,9,EG	-0.5582314	3.2120534
Mode b	1,7,RH	0.0028717	-0.016524
Mode c	7,9,EG	-0.5553562	3.195515
Mismatch			1.46335e-05

Table 2.8: Quasi-resonant triad with two gravity modes coupled with a single Rossby-Haurwitz mode.

Therefore, the efficiency for the RH mode (1,7) is given by equation (2.72) and provided that the Hamiltonian ($\mathcal{H}$) is null, the cubic energy also vanishes, in a way that

$$\begin{aligned} \mathcal{E}_b &= \frac{1}{|A_a|^2 + |A_b|^2 + |A_c|^2} \left(\frac{|\alpha_{a,c}^b|}{|\alpha_{b,c}^a|} |A_a|^2 + \frac{|\alpha_{a,c}^b|}{|\alpha_{a,b}^c|} |A_c|^2 \right) \\ &\leq \max \left\{ \frac{|\alpha_{a,c}^b|}{|\alpha_{b,c}^a|}, \frac{|\alpha_{a,c}^b|}{|\alpha_{a,b}^c|} \right\}, \end{aligned}$$

where the total energy $E_{total}(0)$ was replaced by the kinetic energy. Consequently, the efficiency of energy transfer in the RH mode (1,7) is bounded by

$$\max \left\{ \frac{|\alpha_{a,c}^b|}{|\alpha_{b,c}^a|}, \frac{|\alpha_{a,c}^b|}{|\alpha_{a,b}^c|} \right\} \approx 0.52\%,$$

asserting that the energy centered at this RH mode does not change considerably in time,

although this mode can modulate the period of energy exchange (2.59) among the two gravity modes (cf. RAUPP and DIAS, 2006).

Furthermore, by setting the initial amplitude of mode c as zero and increasing the energy centered initially on mode a, it is possible to ensure an oscillation of almost 100% in both modes a and c as a straightforward consequence of formulas (2.71) and (2.73). Table 2.9 presents the amplitudes considered in three different examples evidencing the role of the RH mode (1,7) in changing the triad's energy transfer period and also the almost steadiness of this mode's kinetic energy. It is worth mentioning that the initial amplitude for the RH mode is chosen as purely imaginary in order to ensure a null Hamiltonian, although the same could be done by distinct amplitudes' phases.

		Example 1	Example 2	Example 3
Mode a	6,9,EG	0.5666241	0.05666241	0.05666241
Mode b	1,7,RH	0.03565586 i	0.03565586 i	0.10696759 i
Mode c	7,9,EG	0.0	0.0	0.0

Table 2.9: Three examples regarding the role of the RH mode in a triad with two gravity modes.

In the first example, the EG mode (6,9) is initialized with a high amplitude, which corresponds to an initial zonal velocity of $200m/s$, while the RH mode (1,7) presents a meridional velocity of $20m/s$. As shown in Table 2.10, the efficiency in the two EG modes achieves more than 99%, meaning that these modes receive almost all the system's energy at different times, and as confirmed by Figure 2.11a, the energy flows just among the EG modes, while the RH mode remains almost stationary during the integration.

		Example 1	Example 2	Example 3
Mode a	6,9,EG	99.6055	71.63432	21.9115
Mode b	1,7,RH	0.51240	0.368513	0.1127
Mode c	7,9,EG	99.09253	71.26535	21.7986

Table 2.10: Efficiencies (%) for each mode in the three examples presented in Table 2.9.

Examples 2 and 3 in Table 2.9 illustrate how the RH mode modulates the period in the triad. In both examples, the EG mode (6,9) has an initial zonal velocity of $20m/s$, while the EG mode (7,9) has a null amplitude. In the second example, the RH mode has initially a zonal velocity of $20m/s$, yielding a vacillation period of more than 12 days, as observed in Figure 2.11b, along with an efficiency of more than 70% in the EG modes. When the energy centered at the RH mode increases, the periods are shortened, and the efficiency may decay as considered in example 3, where the zonal velocity is settled at $60m/s$ for the RH mode. Finally, Table 2.10 also asserts that the kinetic energy in the RH mode never changes more than 0.52% as predicted.

In general, a resonant triad formed by two gravity modes coupled with a RH mode undergoes a similar dynamic to those presented for the triad in Table 2.9, where the RH mode modulates the period of energy exchange, and also, due to its low frequency, and consequently small coupling coefficient, this mode enables an extremely efficient energy

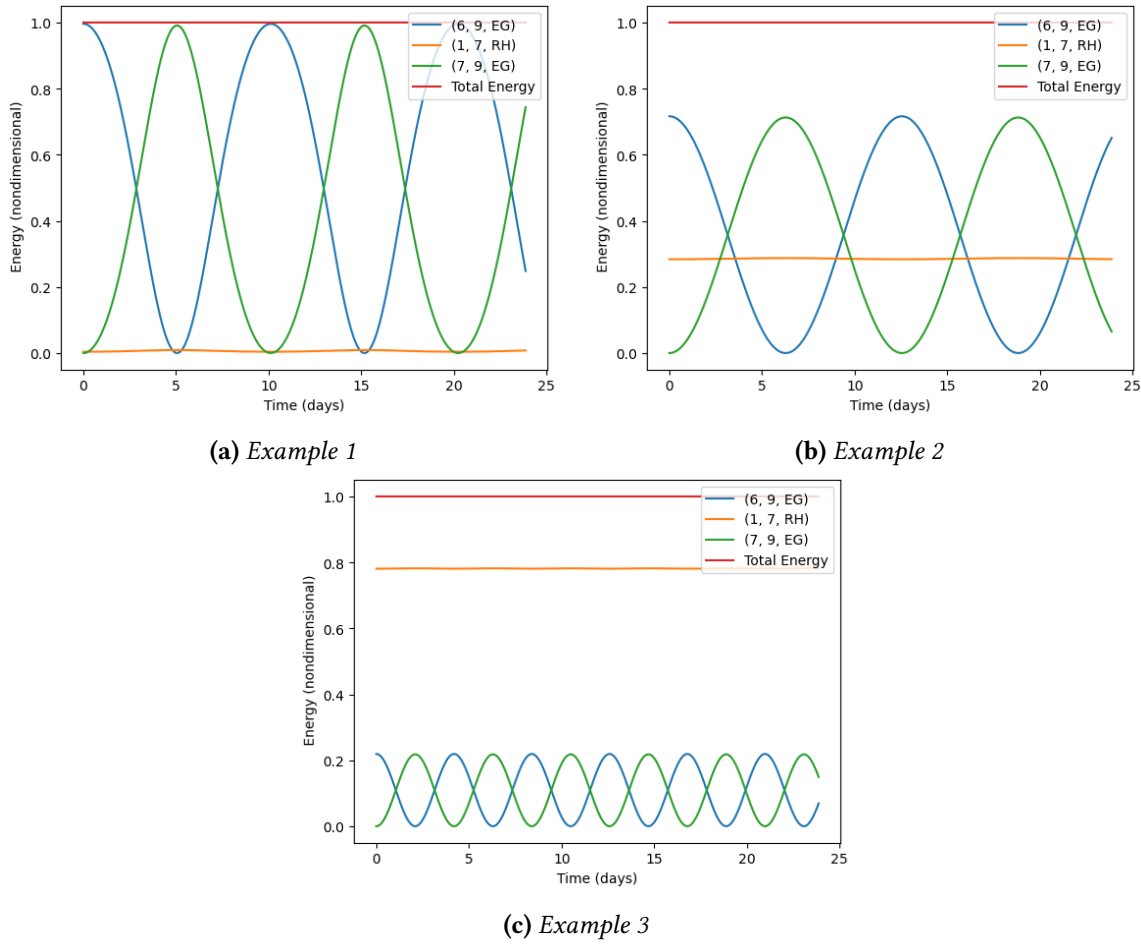


Figure 2.11: Integration of the three examples in Table 2.9 .

transfer among the gravity modes, where all the energy can be transferred from one to another. Even though the RH mode does not change its dynamics through the nonlinear interactions, it plays a central role as the triad's catalyst, where small changes in its initial kinetic energy field alter the amount of energy transferred to the gravity modes (cf. RAUPP and DIAS, 2006).

In fact, considering the triad in Table 2.8 with the EG modes (6,9) and (7,9) having initially 20m/s and zero zonal velocity respectively, and letting the meridional velocity of the RH mode (1,7) vary till 100m/s , one notes, from Figure 2.12a, that the period of energy exchanges is reduced abruptly when the RH mode achieves 20m/s , and the same converges to zero when the kinetic energy centered in the slow mode increases, which does not necessarily mean a faster interaction, since the efficiencies also decrease with the rise of the meridional velocity, as observed in Figure 2.13a. Furthermore, from Figure 2.13a one observes that both EG modes have almost the same efficiency despite the chosen meridional velocity for the RH mode.

Figures 2.12b and 2.13b regard the example in Table 2.4 where the RH mode (2,3) alters substantially the period of energy transfers, acting as a catalyst and decreasing the efficiency of the EG modes when the slow wave has a higher initial energy.

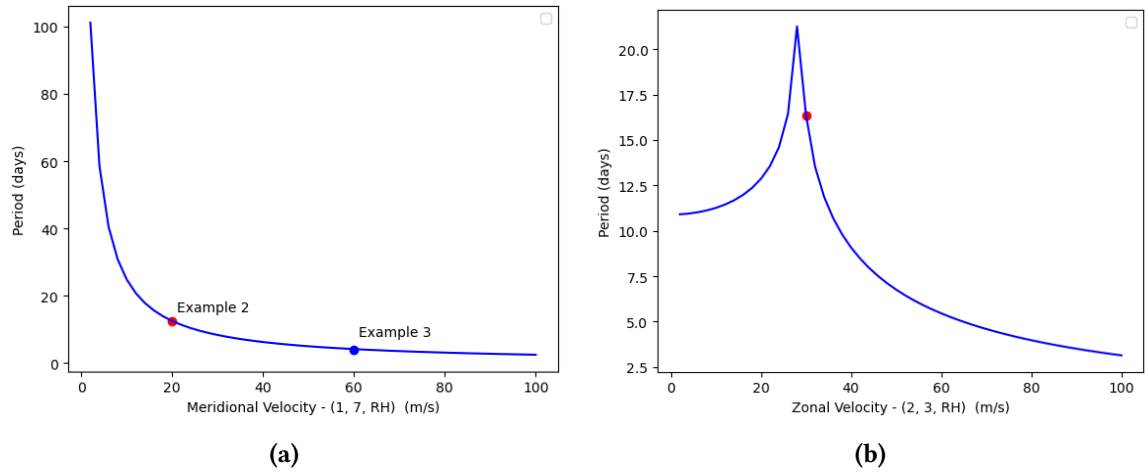


Figure 2.12: Period of the triad when varying the RH modes initial kinetic energy for the Triads in (a) Table 2.9 and (b) Table 2.4.

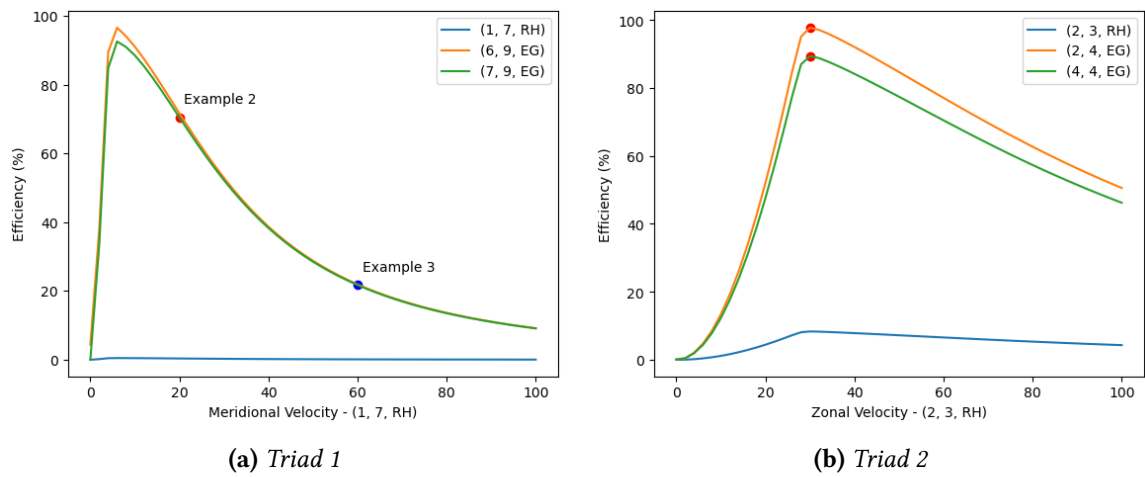


Figure 2.13: Efficiency (%) when varying the RH modes initial kinetic energy for the Triads in (a) Table 2.9 and (b) Table 2.4.

Chapter 3

Coupled Triads

The atmospheric and oceanic dynamics are characterized by a complex network of coupled triads, where a single normal mode can effectively participate in different triads, enabling both direct and inverse energy cascades and the redistribution of energy. Furthermore, when the full partial differential equations are considered, the energy initially centered in a few modes flows toward different waves, which is a possible path to chaos (cf. [LYNCH, 2009](#)).

Triads coupling can essentially occur in a configuration with four or five waves. In the former, two triads are connected by two modes, while in the latter, by a single one, depending on the modes' symmetry. In the equatorial β -plane, [RAUPP and DIAS, 2006](#), considering a five-wave configuration of resonant triads, show that the mode connecting both triads should preferably be the highest frequency mode, such as Inertio-Gravity waves or Kelvin modes. Otherwise, the system could converge to a three-wave system, i.e., two modes are energetically insignificant. In addition to that, the inclusion of two modes increases the *vacillation cycles* in the physical space solution, which is associated with the generation of turbulence.

Although two coupled triads may be seen distant from the full model, this simplification can shed light on the *low frequency variability* of the atmospheric flow, such as the *intraseasonal variability*. [RAUPP and DIAS, 2006](#) suggested that the vacillations enabled by the energy transfers among (two) different resonant waves are related to the observed variability. Considering an adiabatic version of the equatorial β -plane written in isobaric coordinates, [RAUPP, DIAS, et al., 2008](#) also considered the role of Rossby waves in the *tropic-extratropics interactions* on low-frequency time scales, concluding that the energy exchanges among *equatorially trapped first baroclinic equatorial waves* and *barotropic Rossby waves* may be important for the global teleconnection patterns.

As a final remark, [RAPHALDINI et al., 2022](#), adopting both four and five wave configurations, related the efficiency of energy exchanges with the so-called precession resonance (cf. [M. D. BUSTAMANTE et al., 2014](#)), which also enhances the low-frequency oscillation spectrum. Furthermore, the mentioned work suggests that resonant triads might be crucial for the instability of a Rossby mode preferentially excited by forcing mechanisms such as the baroclinic instability.

The triads coupling, in particular configurations using four and five waves, is used here to analyze the barotropic Rossby-Haurwitz modes (1,2) and (4,5) in a more complex triads network. As mentioned in the preceding chapter, the RH modes (1,2) and (4,5) interact mainly with symmetric waves with zonal wavenumber $m = 3$ and meridional wavenumber $n = 4 - 10$, where the energy transfer towards this third wave is less effective as the meridional wavenumber increases. In the following, letting the mentioned RH modes interact simultaneously with the waves (3,4) and (3,6) of the same type, a possible preference for interacting with one of the last modes is analyzed by considering the kinetic energy field of all four modes.

The expansion (2.5) truncated at the four mentioned RH modes yields in equation (2.11) the four-wave system

$$\frac{dA_a}{dt} = -i\omega_a A_a + i\alpha_{b,c}^a A_b A_c + i\alpha_{b,d}^a A_b A_d, \quad (3.1)$$

$$\frac{dA_b}{dt} = -i\omega_b A_b + i\alpha_{a,c}^b A_a A_c^* + i\alpha_{a,d}^b A_a A_d^*, \quad (3.2)$$

$$\frac{dA_c}{dt} = -i\omega_c A_c + i\alpha_{a,b}^c A_a A_b^*, \quad (3.3)$$

$$\frac{dA_d}{dt} = -i\omega_d A_d + i\alpha_{a,b}^d A_a A_b^*, \quad (3.4)$$

whose *Manley-Rowe* relations, which are preserved during integration, are

$$J_{a,c,d} = |A_a|^2 + \frac{\alpha_{b,c}^a}{\alpha_{a,b}^c} |A_c|^2 + \frac{\alpha_{b,d}^a}{\alpha_{a,b}^d} |A_d|^2,$$

$$J_{b,c,d} = |A_b|^2 - \frac{\alpha_{a,c}^b}{\alpha_{a,b}^c} |A_c|^2 - \frac{\alpha_{a,d}^b}{\alpha_{a,b}^d} |A_d|^2.$$

It is important to mention that even though the system (3.1) - (3.4) was derived from the RSW equations, whose total energy is given by the Hamiltonian (1.27), the four-wave system *does not necessarily conserve* the total energy obtained from the Hamiltonian calculated at the truncated solution (see Appendix). For models whose total energy is quadratic (e.g., barotropic nondivergent model), this amount is conserved for *any* truncation (cf. MAJDA and WANG, 2006).

Therefore, there exist Galerkin truncations (2.5) that are not consistent with the system, in the sense that the total energy may not be conserved and, consequently, the analysis performed for the isolated triads in the previous chapter is no longer adequate, since the definition of efficiency (2.36) depends on a division by the total energy, which is not preserved. Hence, the percentage of total energy is not well defined during time, in such a way that the efficiency is not a good measure to compare quartets.

One way to avoid division by the total energy is to consider the absolute kinetic energy in Joules of each mode, which is given simply by

$$EK_a = (gh_0^2 a^2 \pi) \cdot \rho \cdot \|A_a\|^2, \quad (3.5)$$

where ρ is the *density of air*. This formula is a direct consequence of the Hamiltonian (1.26) where the scales are considered. Using the kinetic energy, the percent increase (or decrease) of this energy can be measured by comparing the four-wave configuration and the triad using the simple formula

$$\mathcal{P}_a = \frac{\Delta EK_a^4(t) - \Delta EK_a^3(t)}{\Delta EK_a^3(t)}, \quad (3.6)$$

where EK_a^4 represents the kinetic energy of the mode a in the quartet, while EK_a^3 in the triad, which is chosen as the triad with only RH modes, as will be explained below, and $\Delta EK = \max_t EK - \min_t EK$ is the total variation of the kinetic field in each mode.

Hereafter, the dimensionless energy corresponds to just the term $\|A_a\|^2$ in (3.5), i.e., in the following graphs the absolute value of the amplitudes is being considered, instead of the normalization by the initial total energy, as in the case for three waves.

After these remarks, a first example regarding the interaction of the RH modes (4,5) and (1,2) with the slow waves (3,4) and (3,6) is considered in Table 3.1, where the zonal velocities of the last two modes are initially chosen as zero in order to analyze how the energy centered at the first modes is redistributed to the other two, leading to a possible preference for one mode.

		Freq.	Coeff. 1	Coeff. 2	Zonal	Amp
Mode a	4,5,RH	-0.12522	0.3638	-0.5516	40	0.122418
Mode b	1,2,RH	-0.09934	-0.0284	-0.2082	40	0.0927075
Mode c	3,4,RH	-0.13431	0.4434	-	0	0
Mode d	3,6,RH	-0.065826	-	-0.343	0	0

Table 3.1: Four-wave configuration containing both RH modes (4,5) and (1,2).

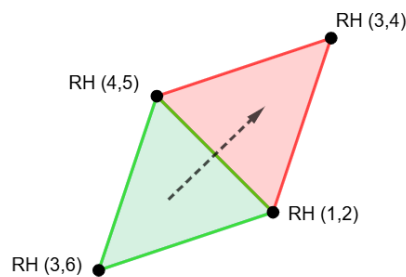


Figure 3.1: Schematic representation of the quartet in Table 3.1.

As exemplified in Figure 3.2, there is a notorious preference of the RH modes (4,5) and (1,2) to interact with the RH mode (3,4), a fact characterized by the higher peaks in the kinetic energy field of the last mode when compared to the same field of the RH mode (3,6). One observes that the three RH modes with the smallest zonal wavenumbers present a quasi-period behavior, and the period of the RH mode (3,4) is almost half the period of the other two modes, which is a possible explanation for the higher increase

in the kinetic energy field of this mode. Furthermore, the RH mode (4,5) presents a less regular behavior, where there is not a determined period for the moments of receiving and transmitting energy to this mode, in a manner that this four-wave system shows how nonlinear interactions among waves can lead to a turbulent system, in contrast to the periodicity of triadic systems.

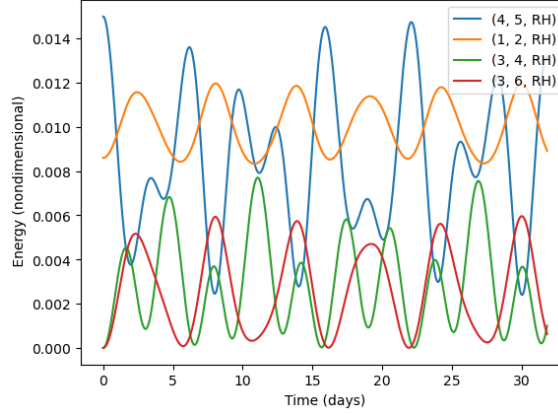


Figure 3.2: Integration of the four-wave configuration presented in Table 3.1. In the y-axis is considered the squared absolute value of the amplitudes, corresponding to the nondimensional form of the kinetic energy EK .

When the two coupled triads are considered separately, Figure 3.3 shows that the period of the triad containing the RH mode (3,4) is shorter than the period of the other triad. Furthermore, the kinetic energy in the RH mode (3,4) *increases* more than 35% (i.e., $\mathcal{P}_{(3,4)} = +0.35$) in the four-wave model, while the same energy centered in the RH mode (3,6) decreases more than 26% (i.e., $\mathcal{P}_{(3,6)} = -0.26$), which can be partially explained by the difference in the periods, where the faster exchanges towards mode (3,4) increase their energy. The kinetic energy in the RH mode (1,2) suffers the influence of two opposite mechanisms: a stronger flux due to the presence of the RH mode (3,6) as considered in Figure 3.3a and an almost stationary pattern originated by the RH mode (3,4) as plotted in Figure 3.3b. These two distinct exchanges can explain why the kinetic energy in wave (1,2) tends to the average of the energies obtained in each triad.

The periods of energy exchange of the different triads appear to play a central role in energy redistribution. In fact, triads with shorter periods tend to contain a more expressive fraction of the total energy because the energy flows faster among these modes. The coupled triads in Table 3.1 present this suggestion, where the second triad, containing the RH mode (3,4) as its pump, has a shorter period than the first triad, enabling a higher and more effective energy transfer toward mode (3,4). In a more dramatic way, when the RH mode (3,6) is replaced by the RH mode (3,10) in Table 3.1, the period of energy exchanges for the new mode increases to more than 297 days and the difference between this period and the one for mode (3,4) leads the four-wave configuration to a three-wave system (cf. RAUPP and DIAS, 2006).

Figure 3.4 shows the integration for the triad comprising the RH modes (1,2), (3,10), and (4,5), where the kinetic energy of mode (3,10) is not constant and a higher peak is observed in mode (1,2), which doubled its initial kinetic energy. However, since the exchanges

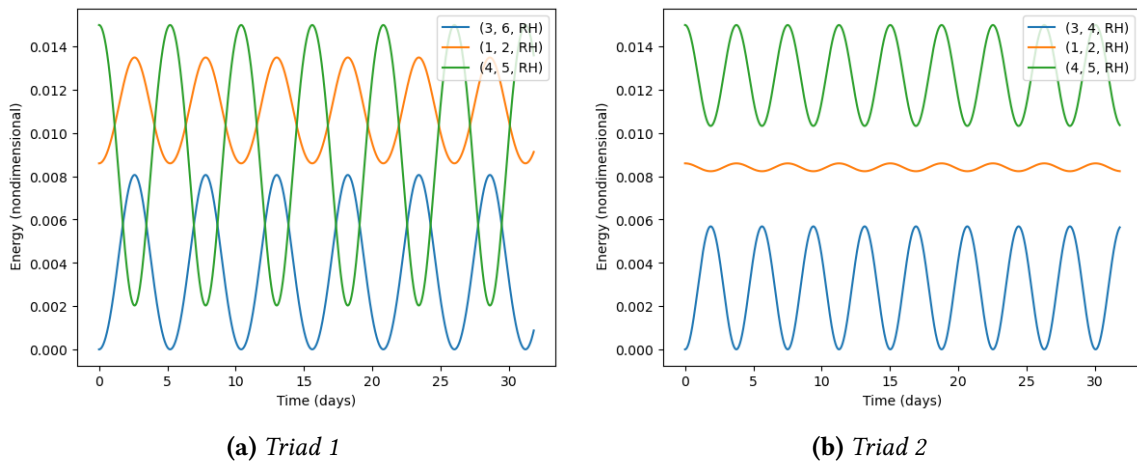


Figure 3.3: Integration of the two triads corresponding to the four-wave configuration presented in Table 3.1. (a) Triad completed with RH mode (3,6) (b) Triad completed with RH mode (3,4).

are slower, presenting a high period, the faster dynamics of the triad containing mode (3,4) dominates the system, and mode (3,10) becomes unexpressive and almost steady, as observed in Figure 3.5. Furthermore, mode (3,10) presenting irrelevant dynamics alters directly the kinetic energy field of wave (1,2), in which the observed peak is no longer existent, and the energy is comprised in a smaller variation.

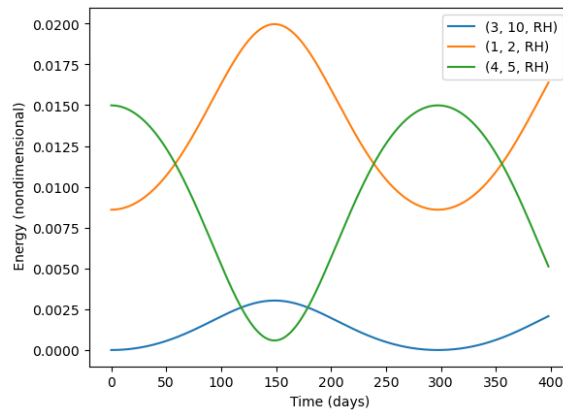


Figure 3.4: Integration of the triad containing just RH modes in Table 3.1 with the RH mode (3,6) replaced by the (3,10).

These examples suggest that a stronger transfer is obtained by a balance between *period* and *initial energy*, where triads with lower periods and higher initial energy can receive more energy, resulting in a higher energy centered in each mode of the considered triad.

In the following, the central question of this work is considered: how does a gravity wave alter the dynamics of RH modes in terms of periods and energy exchanges.

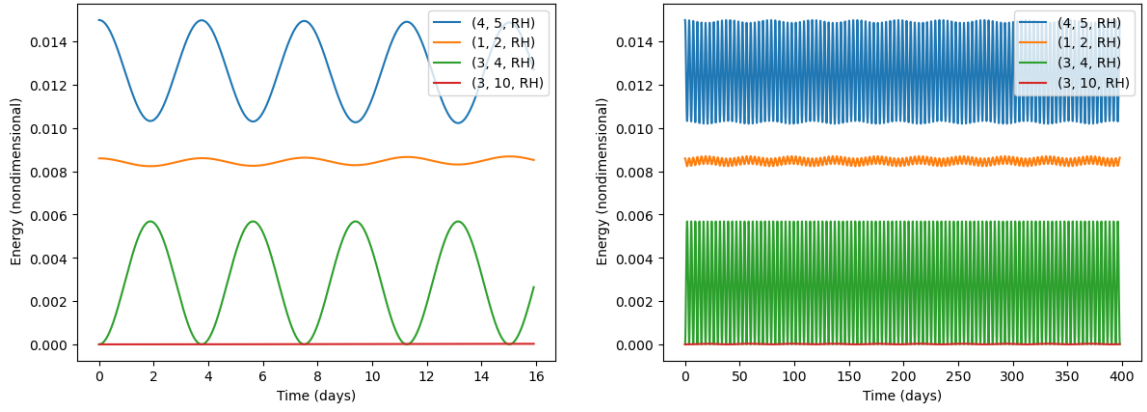


Figure 3.5: Integration during different times for the four-wave configuration as in Table 3.1 with the RH mode (3,6) replaced by the (3,10).

3.1 Inertia-Gravity Waves

Although gravity waves are fast waves with higher frequencies, gravity modes, in particular Kelvin waves, are associated with distinct meteorological phenomena, such as the aforementioned Madden-Julian Oscillation (cf. [TERUYA, RAPHALDINI, and RAUPP, 2022](#)) and intraseasonal variability (cf. [RAUPP and DIAS, 2006](#)). As shown by [RAUPP, TERUYA, et al., 2019](#), inertio-gravity waves may also act as a catalyst mode for acoustic waves, controlling both the period and efficiency of the energy exchanges among the acoustic modes, resembling the role of the RH modes in the RSW model, as considered in Figure 2.11. Furthermore, using reanalysis data, [TERUYA, RAPHALDINI, MAYTA, et al., 2023](#) obtained the expected dominant global-scale Rossby and Kelvin waves and also unexpected events such as the eastward propagation of westward-gravity waves, possibly due to a strong coupling with Kelvin waves.

Therefore, gravity waves play an important role in atmospheric and oceanic dynamics, being determinant in the generation of energy cascade processes, which are associated, for example, with the dissipation of oceanic oscillations by the inverse cascade of Rossby waves (cf. [ROCHA et al., 2018](#)).

When triads are coupled together, gravity waves may become catalyst modes for RH waves, generating stronger fluxes of energy toward these modes. In fact, considering a triad of RH modes coupled with a single gravity mode, the presence of the gravity mode can alter the energy exchanges mainly in two ways: either the interactions among RH waves are attenuated, even though the gravity mode does not receive a considerable amount of energy, or the RH modes belonging to both two triads in this four-wave configuration have a stronger oscillation, where the energy from the gravity mode is summed up to the energy from the remaining RH mode. In this sense, gravity waves change the RH wave dynamics, and once they are filtered out from the system, some mechanisms that mitigate or enhance the transfer among RH waves are disregarded, which can possibly have important consequences in the weather forecast.

In order to illustrate the role of gravity modes in altering the dynamics of RH modes, the four-wave configuration presented in Table 3.2 is considered. The following three

examples were chosen to analyze distinct scenarios. In the first, the eastward gravity mode (1,1) is initialized without kinetic energy, i.e., its initial velocity is set to zero to compare the distribution of the same amount of initial energy with and without the gravity mode. The second example considers the case where the gravity mode initially has the highest kinetic energy, while the third example is the opposite, when a RH mode has the highest energy at the beginning. These three cases along with the initial zonal velocities and amplitudes chosen are shown in Table 3.3.

		Freq.	Coeff. 1	Coeff. 2
Mode a	4,5,RH	-0.12522	0.3638	-0.5572
Mode b	3,4,RH	-0.13431	0.4434	-0.5902
Mode c	1,2,RH	-0.09934	-0.0284	-
Mode d	1,1,EG	0.36913	-	-0.1323

Table 3.2: Four-wave configuration to study the influence of gravity waves in the dynamics of Rossby-Haurwitz modes.

	Example 1		Example 2		Example 3	
Mode	Zonal	Amplitude	Zonal	Amplitude	Zonal	Amplitude
4,5,RH	30	0.0918137	30	0.0918137	30	0.122418
3,4,RH	30	0.0836648	30	0.0836648	30	0.111553
1,2,RH	30	0.0695306	30	0.0695306	30	0.0695306
1,1,EG	0	0	80	0.1905199	20	0.04763

Table 3.3: Examples regarding how a single EG mode can affect the dynamics of three RH modes.

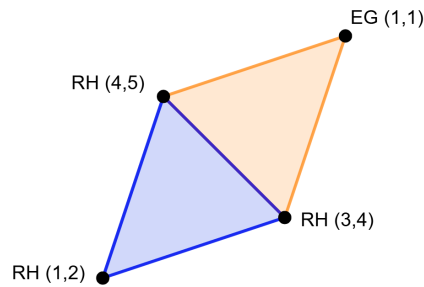


Figure 3.6: Schematic representation of the quartet in Table 3.2.

In the first example, the eastward gravity mode (1,1) has a zero initial amplitude, and consequently, the energy is initially distributed among the RH modes, which is a common choice when considering the PDE model, since the energy is projected into the chosen modes to observe how the energy flows among the network of normal modes and which are the modes excited by the primary ones (cf. [RAPHALDINI et al., 2022](#)).

As observed in Figure 3.7b, the EG mode (1,1) does not allow significant energy exchanges in the triad formed with the RH modes (4,5) and (3,4), in a way that the amplitudes of the modes stay almost constant during integration. This fact is an immediate consequence of the chosen initial amplitudes (cf. [RAUPP and DIAS, 2006](#)), as the same triad was

considered in Figure 2.7 with distinct initial conditions, yielding an efficiency of more than 9% in the EG mode. On the other hand, as shown in Figure 3.7a, the three RH modes interact generating modulations undergoing periods of more than 4 days, where the RH mode (1,2) acts just as a catalyst for the other two modes.

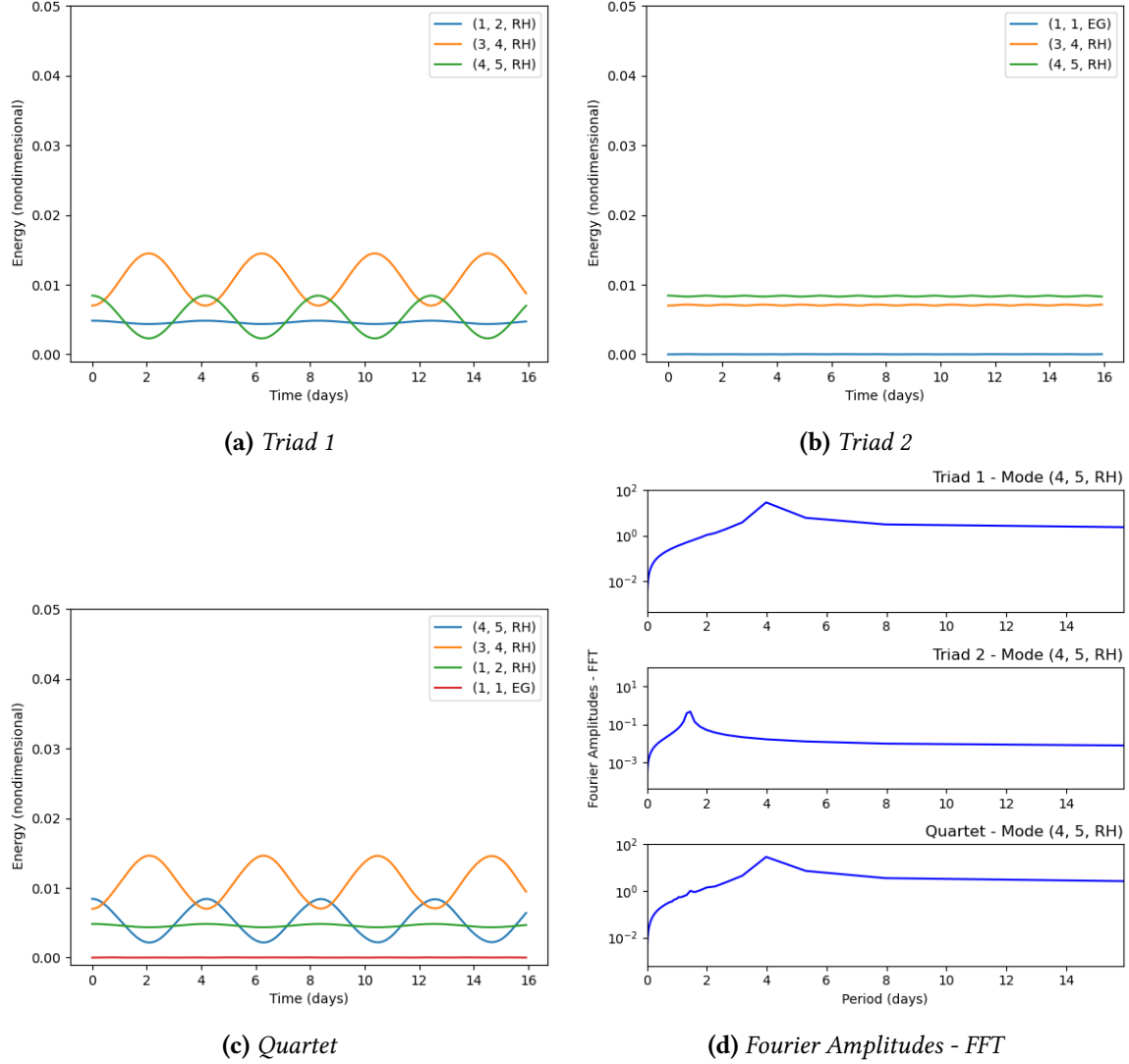


Figure 3.7: Integration of the (a) first triad (b) second triad (c) four-wave configuration in Table 3.3 example 1. The graph in (d) corresponds to the Fourier periods for the RH mode (4,5) obtained via FFT of the time series representing the kinetic energy field of mode (4,5).

Letting the four waves interact non-linearly, Figure 3.7c shows that the EG mode (1,1) is not affected by the other three RH modes, and the RH mode (1,2) has a small variation in its kinetic energy. The EG wave (1,1) does not change the system in this case, and the dynamics of the four-wave configuration is exactly the same of the triad containing only RH modes.

Proceeding with the Fourier analysis of the functions representing the kinetic energy of the modes and in particular, analyzing the dominant frequencies (periods) of the four-wave configuration, one arrives at Figure 3.7d, where the periods along with the corresponding

Fourier amplitudes are shown. One observes that the peak in the quartet is the same obtained for the triad formed by just RH modes, while, the peak regarding the influence of the EG mode (1,1), being around 1.6 days, has a lower Fourier amplitude in the quartet, and consequently its influence is less significant.

The second example in Table 3.3 initializes the EG mode (1,1) with a faster zonal velocity, which is the common situation in the real atmosphere since gravity modes have higher frequencies and therefore a shorter period of oscillation. When more energy is initially centered on the EG mode, the RH modes in its triad interact more efficiently, undergoing visible periods of maximums and minimums in the kinetic energy field, as readily seen in Figure 3.8b. Unlike the first example, here the presence of the EG mode favors the transference of energy toward the RH modes (4,5) and (3,4). In fact, Table 3.4 shows that the previously defined percent $\mathcal{P}$ is considerably positive for modes (4,5) and (3,4), which means that the observed variations in the quartet are higher, and consequently the energy exchanges are stronger, where a higher amount of energy flows through these waves. For the calculus of $\mathcal{P}$ for the RH modes (3,4) and (4,5) the variations of the triad of RH modes (ΔEK_1^3) and the quartet (ΔEK^4) were used, while for the other modes the two existing variations were considered. Furthermore, the RH mode (1,2) and the EG mode (1,1) do not receive the same amount of energy as in their triads, which is characterized by the negative $\mathcal{P}$, i.e., while the interactions between the RH modes (3,4) and (4,5) become stronger, the role of the other two modes is softened.

The Fourier amplitudes in Figure 3.8d also reveal that the period in the quartet has become longer, around 8 days, although another dominant period around 1.6 days due to the gravity wave is also observed. Consequently, the presence of a gravity wave generates a longer period and stronger interactions, significantly altering the dynamics of RH modes.

Mode	ΔEK_1^3	ΔEK_2^3	ΔEK^4	$\mathcal{P}(\%)$
4,5,RH	0.006136	0.007142	0.008426	37.3115
3,4,RH	0.007478	0.00757	0.009549	27.70768
1,2,RH	0.0004796	-	0.000307	-35.9831
1,1,EG	-	0.0016962	0.001604	-5.43584

Table 3.4: Energy variations and $\mathcal{P}$ -measure (%) for the waves in Table 3.3 in the second example.

Finally, the third example in Table 3.3 corresponds to a case where the presence of the EG mode also increases the kinetic energy of the RH modes (3,4) and (4,5). The kinetic energy fields are shown in Figure 3.9 for both triads together with the fields for the four-wave configuration. Table 3.5 highlights that the RH modes (3,4) and (4,5) receive more kinetic energy in the quartet than in the triad of RH waves and even though the variation in the energy field of the gravity wave is discreet, the last wave enables stronger interactions between the former RH modes. Furthermore, as presented in Figure 3.9d, the dominant frequencies for the four-wave configurations are the combination of the frequencies of the single triads, where the faster oscillations around 1.7 days are observed in the triad containing the gravity mode, while the slower vacillation around 4 days regards the influence of the RH mode (1,2). In this case, the periods of the triads were followed, in the sense that longer (or shorter) periods were not generated.

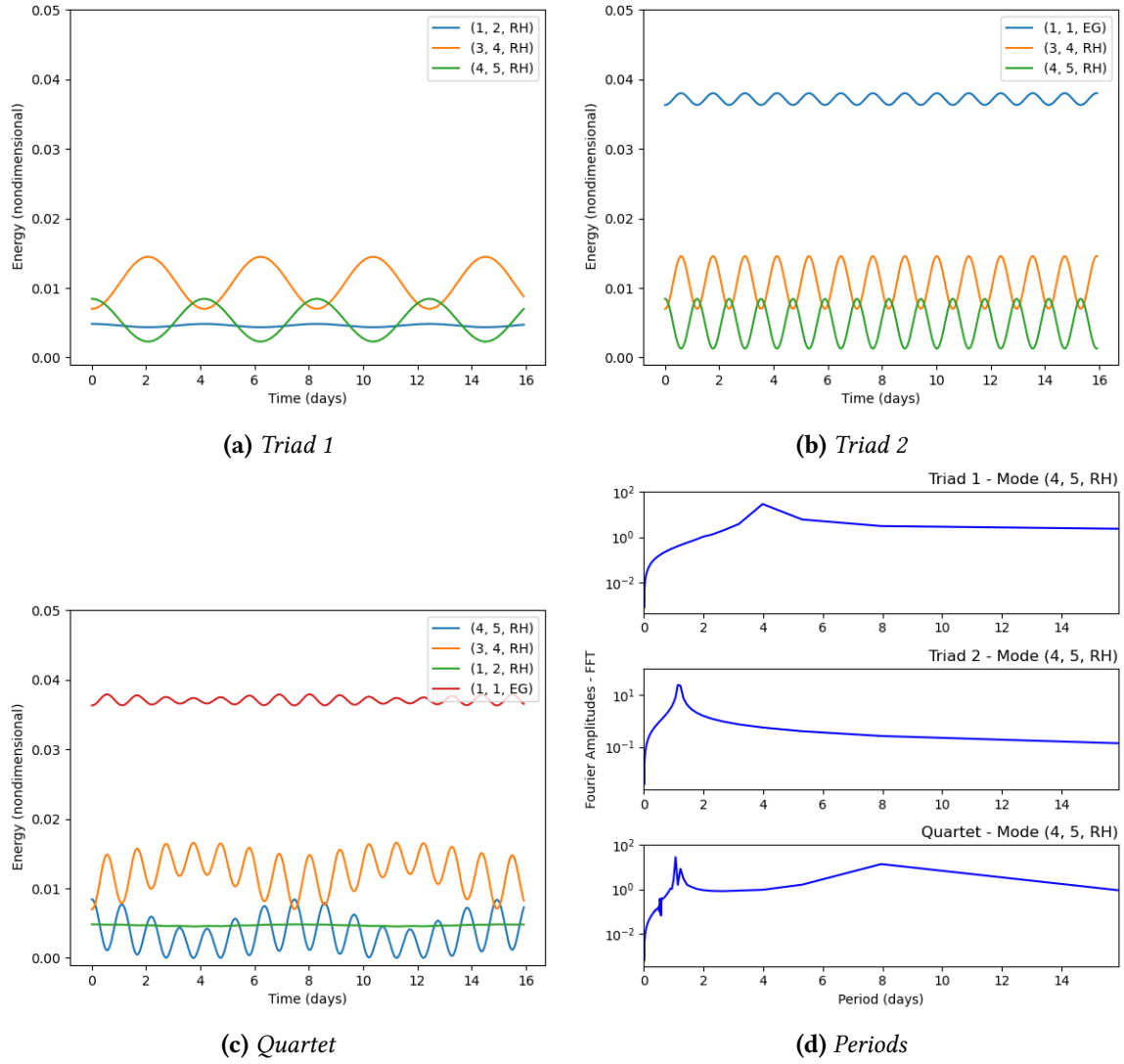


Figure 3.8: Integration of the (a) first triad (b) second triad (c) four-wave configuration in Table 3.3 example 2. The graph in (d) corresponds to the Fourier periods for the RH mode (4,5) obtained via FFT of the time series representing the kinetic energy field of mode (4,5).

Mode	ΔEK_1^3	ΔEK_2^3	ΔEK^4	$\mathcal{P}(\%)$
4,5,RH	0.006166	0.002383	0.007716	25.74323
3,4,RH	0.007478	0.002524	0.009099	21.67822
1,2,RH	0.0004796	-	0.000468	-2.409459
1,1,EG	-	0.000566	0.0005682	0.406648

Table 3.5: Energy variations and $\mathcal{P}$ -measure (%) for the waves in Table 3.3 in the third example

The presence of a gravity wave interacting with RH modes can make the energy transfer stronger or weaker, and this interference can be measured by the percentage $\mathcal{P}$ as defined in (3.6), where the percentage increase in the variation of the kinetic energy is observed, which corresponds to the amount of extra kinetic energy flowing toward the studied mode. With the end of considering conditions to increase (or decrease) the

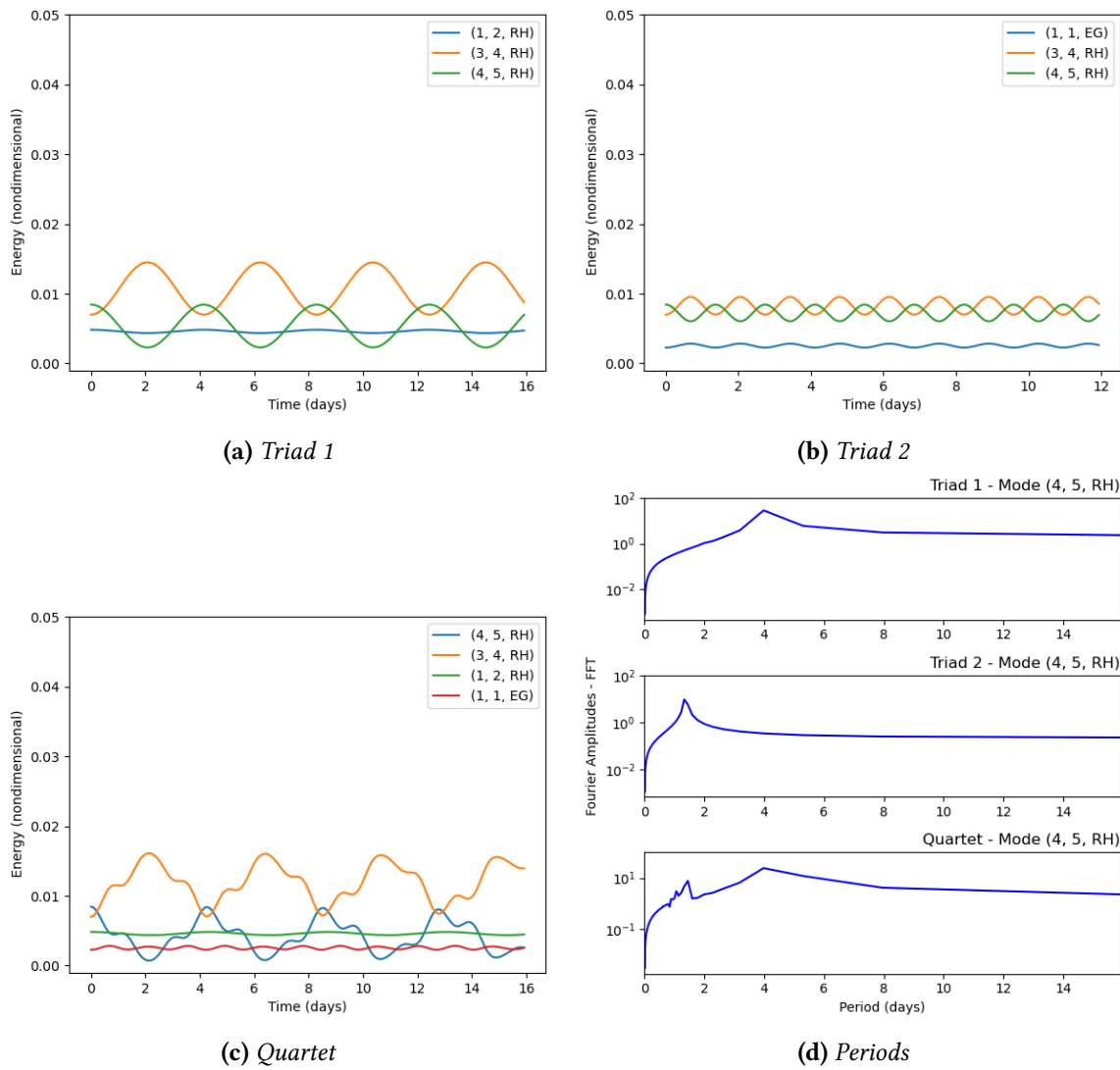


Figure 3.9: Integration of the (a) first triad (b) second triad (c) four-wave configuration in Table 3.3 example 3. The graph in (d) corresponds to the Fourier periods for the RH mode (4,5) obtained via FFT of the time series representing the kinetic energy field of mode (4,5).

variation of kinetic energy in RH modes, Figure 3.10 generalizes the three presented examples by fixing the zonal velocity of the RH modes (4,5) and (3,4) at 30m/s and letting the zonal velocity of both RH mode (1,2) and EG mode (1,1) vary between 0 and 100m/s. In this figure, the measure $\mathcal{P}$ was bounded by 1, in a manner that values higher than 1 are represented in the same region as this maximum value. The three examples in Table 3.3 are displayed on the dotted line. One notes that, increasing the zonal velocity of the RH mode (1,2), the immediate effect is the reduction of the variation, which can become negative, i.e., the gravity wave inhibits the energy transfer between RH waves, in such a way that less energy is centered at RH modes. In fact, the blue regions correspond to triads of RH modes with stronger interactions, due to the higher velocities of mode (1,2), and therefore, the measure $\mathcal{P}$ tends to decrease in these scenarios. Furthermore, Figure 3.10 indicates regions (white) where the presence of the EG mode (1,1) does not change the kinetic energy of the RH modes (3,4) and (4,5), where the interactions are reduced to

the case of three RH waves.

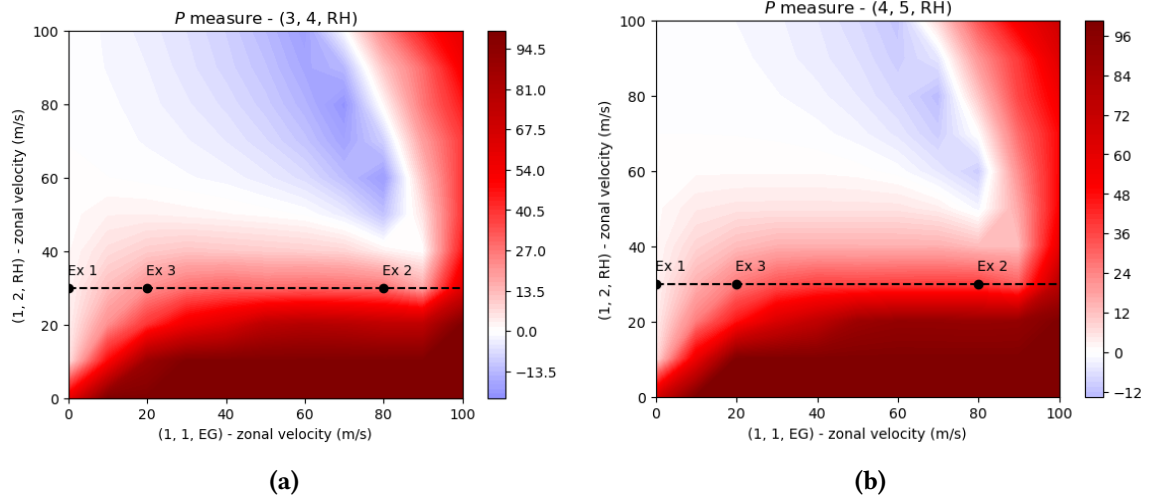


Figure 3.10: $\mathcal{P}$ -measure (%) for the RH modes (3,4) and (4,5) initially with zonal velocity 30m/s in the quartet presented in Table 3.2. Here, a grid of 50×50 points was considered.

Figure 3.11 shows a profile of the $\mathcal{P}$ measure graphs where the zonal velocity of the RH mode (1,2) is fixed at 30m/s , corresponding to the examples presented in Table 3.3. One observes that for all velocities the measure is positive for the RH modes (3,4) and (4,5), and therefore the amount of kinetic energy received by these modes increases in the four-wave configuration. This increase achieves a plateau when the zonal velocity of the EG mode is between 30 and 90m/s , and since the energy in the triad of RH modes does not change (the initial velocity is constant for these examples), the kinetic energy of the RH modes is almost equally influenced by different velocities of the gravity wave. Furthermore, there is a break near 90m/s and a peak near 130m/s , which can possibly be related to the periods and the chosen initial energy field, in a way that the precession resonance could play a relevant role in generating these peaks (cf. [M. D. BUSTAMANTE et al., 2014](#)).

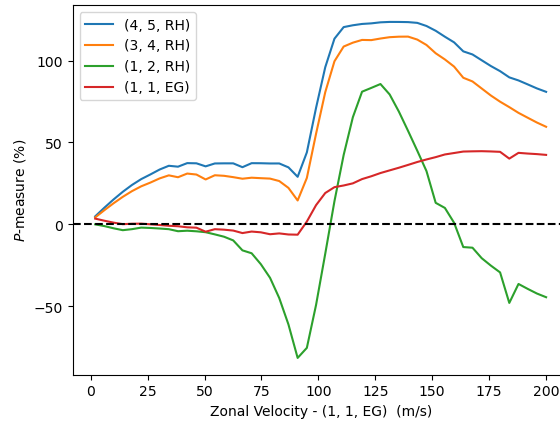


Figure 3.11: $\mathcal{P}$ -measure for the modes in the quartet presented in Table 3.2, with fixed zonal velocity of 30m/s for each RH mode.

The examples previously considered highlighted how a gravity wave can change

not only the kinetic energy field but also the period of energy transfers, in which case the exchanges become slower, even though the period of energy transfers in the triad containing the EG mode was important, being one of the dominant periods in the quartet. Figure 3.12a shows the difference between the longer dominant period, i.e., the largest-period local maximum in absolute value observed in the Fourier frequency spectrum (as in Figures 3.7d, 3.8d, and 3.9d), and the period of the triad containing only the RH modes, when varying the initial zonal velocities of both the RH mode (1,2) and the EG mode (1,1). One notes that the difference is always positive, the larger period never decreases, and the majority of velocity combinations lead to an increment of less than 6 days in the longer period.

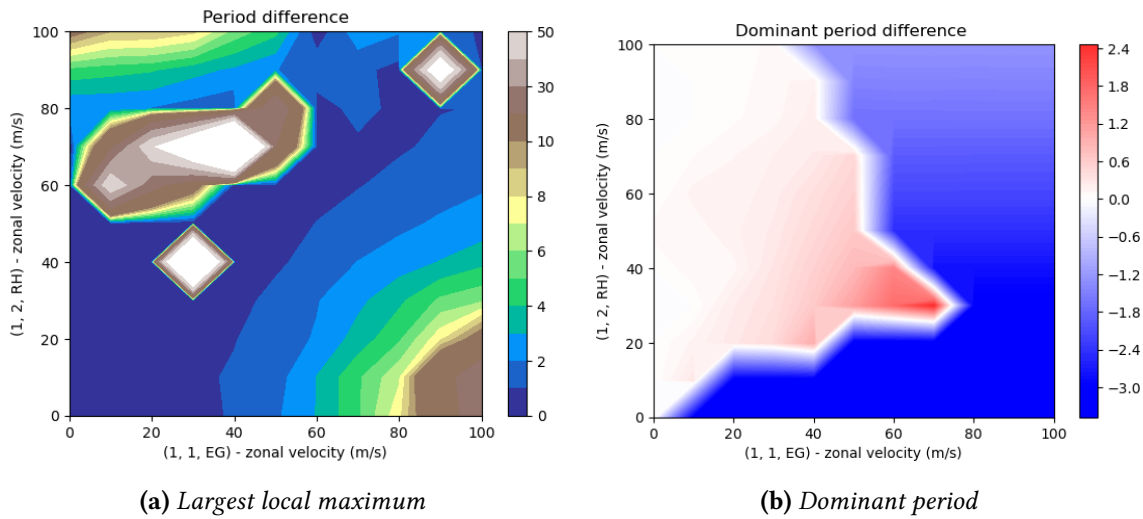


Figure 3.12: (a) The difference in days between the largest-period local maximum in the Fourier spectrum for the four-wave configuration and the triad of RH modes. (b) The difference in days between the dominant periods. Here, a grid of 50×50 points was considered.

Furthermore, there are cases where the period increases more than 50 days, which are represented by the white regions. In these combinations, the period of the triad containing the EG mode remains relevant in the quartet, even though a longer oscillation period appears. As an example, consider the center of the lower white diamond in Figure 3.12a, which corresponds to a zonal velocity of around 40 m/s for the RH mode (1,2), while 36 m/s for the gravity mode. As observed in Figure 3.13a, a longer period of around 80 days is obtained, although the faster oscillations of around 3 days characterizing the triad containing the EG mode (Figure 3.13b) is also present in the dynamics of the quartet.

Finally, Figure 3.12b asserts the difference in the dominant period, i.e., the period with the highest Fourier amplitude, between the four-wave configuration and the triad of RH modes. The observed reductions and increases are around 3 days, where the regions with stronger reductions correspond to either low energy centered at the RH mode (1,2) or higher energy at the gravity mode, and consequently, the triad of RH modes has a softer interaction, while the quartet presents periods of faster transfer due to the gravity wave, characterizing the obtained reduction.

Although the EG mode (1,1) is important when modeling the Madden-Julian Oscillation (cf. TERUYA, RAPHALDINI, RAUPP, *et al.*, 2024), normal modes with low zonal wavenumber

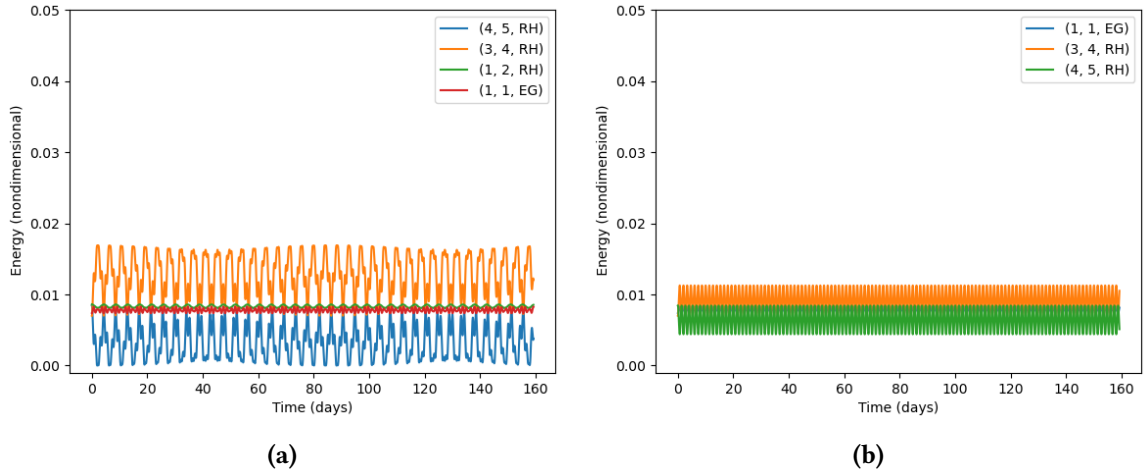


Figure 3.13: Integration of the (a) four-wave configuration in Table 3.2 (b) the triad of RH modes, where the modes have initial zonal velocities respectively given by 30m/s, 30m/s, 40m/s, 36m/s.

are well represented in weather forecast models, and more drastic consequences when filtering out gravity modes should be observed for waves with higher zonal wavenumbers, whose representation is not satisfactory due to the high velocities encountered in these waves. In order to study gravity waves with higher frequencies, the quartet formed by the previous three RH modes joined with the EG mode (7,9) is considered, and the kinetic energy analysis is pursued. Table 3.6 presents the coupling coefficients and frequencies for the quartet, including the EG mode (7,9).

		Freq.	Coeff. 1	Coeff. 2
Mode a	4,5,RH	-0.12522	0.3638	0.7652
Mode b	3,4,RH	-0.13431	0.4434	0.8986
Mode c	1,2,RH	-0.09934	-0.0284	-
Mode d	7,9,EG	3.19115	-	0.33802

Table 3.6: Second four-wave configuration to study the influence of gravity waves in the dynamics of Rossby-Haurwitz modes.

As before, varying the initial amplitudes for the waves participating in a single triad, i.e., the RH mode (1,2) and the EG mode (7,9), it is possible to observe the variation differences ($\mathcal{P}$ -measure) in the modes linking both triads. As considered in Figure 3.14, the presence of the EG mode always increases the kinetic energy, where stronger fluxes are obtained with velocities such that in the cases of the EG mode (1,1) a reduction in the kinetic energy was observed (see Figure 3.10). Although increasing the energy centered on the RH modes (3,4) and (4,5), the EG mode does not receive energy, and its kinetic energy field stays constant during integration. Furthermore, there is no significant change in the dominant periods by the presence of the gravity wave (7,9).

The previous two examples on the influence of EG modes in altering the dynamics of RH modes highlighted how the kinetic energy fields and periods of energy exchange can be drastically changed. In the next section, both EG waves (1,1) and (7,9) are considered together, and the same analysis of energy and period is performed to study whether the

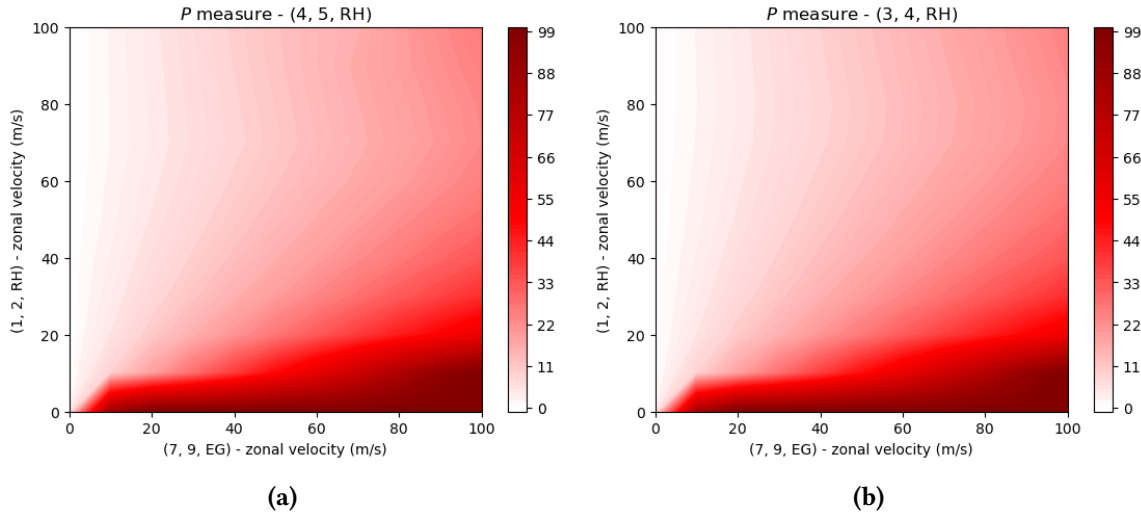


Figure 3.14: $\mathcal{P}$ -measure (%) for the RH modes (3,4) and (4,5) initially with zonal velocity 30m/s in the four-wave model presented in Table 3.6. Here, a grid of 50×50 points was considered.

difference increases negatively or if there is some balance between the EG modes.

3.2 Five-Wave model

A more complex system can be obtained from the four-wave configurations considered above by combining them. In this case, the arrangement of the five modes generates three triads (see Figure 3.15), where the RH modes (3,4) and (4,5) belong to all of them. When the Galerkin expansion (2.5) is truncated into three RH modes (4,5), (3,4) and (1,2), along with the eastward gravity waves (1,1) and (7,9), which satisfy the zonal wavenumber conditions

$$m_a = m_b + m_c,$$

$$m_a = m_b + m_d,$$

$$m_e = m_a + m_b,$$

the nonlinear interactions are expressed as the five-equation system

$$\frac{dA_a}{dt} = -i\omega_a A_a + i\alpha_{b,c}^a A_b A_c + i\alpha_{b,d}^a A_b A_d + i\alpha_{b,e}^a A_b^* A_e, \quad (3.7)$$

$$\frac{dA_b}{dt} = -i\omega_b A_b + i\alpha_{a,c}^b A_a A_c^* + i\alpha_{a,d}^b A_a A_d^* + i\alpha_{a,e}^b A_a^* A_e, \quad (3.8)$$

$$\frac{dA_c}{dt} = -i\omega_c A_c + i\alpha_{a,b}^c A_a A_b^*, \quad (3.9)$$

$$\frac{dA_d}{dt} = -i\omega_d A_d + i\alpha_{a,b}^d A_a A_b^*, \quad (3.10)$$

$$\frac{dA_e}{dt} = -i\omega_e A_e + i\alpha_{a,b}^e A_a A_b, \quad (3.11)$$

where the coupling coefficients are as in (2.19) - (2.21).

Considering the initial zonal velocity of the three RH modes (4,5), (3,4) and (1,2) fixed at 30m/s and varying the zonal velocity of the EG modes (1,1) and (7,9), one obtains the graphs displayed in Figure 3.16, where it is readily seen that the combination of both EG waves yields stronger fluxes of energy towards the RH modes (3,4) and (4,5). There are no longer regions of negative $\mathcal{P}$ -measure, and consequently the kinetic energy variations always increase in the mentioned RH waves. Furthermore, in the majority of the possible combinations of zonal velocities, the kinetic energy increases more than 25%, which is a considerable amount since the kinetic energy in the real atmosphere is in the order of 10^{21}J .

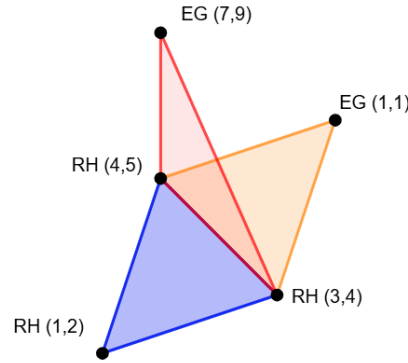


Figure 3.15: Five-wave configuration with three RH modes coupled with two EG modes.

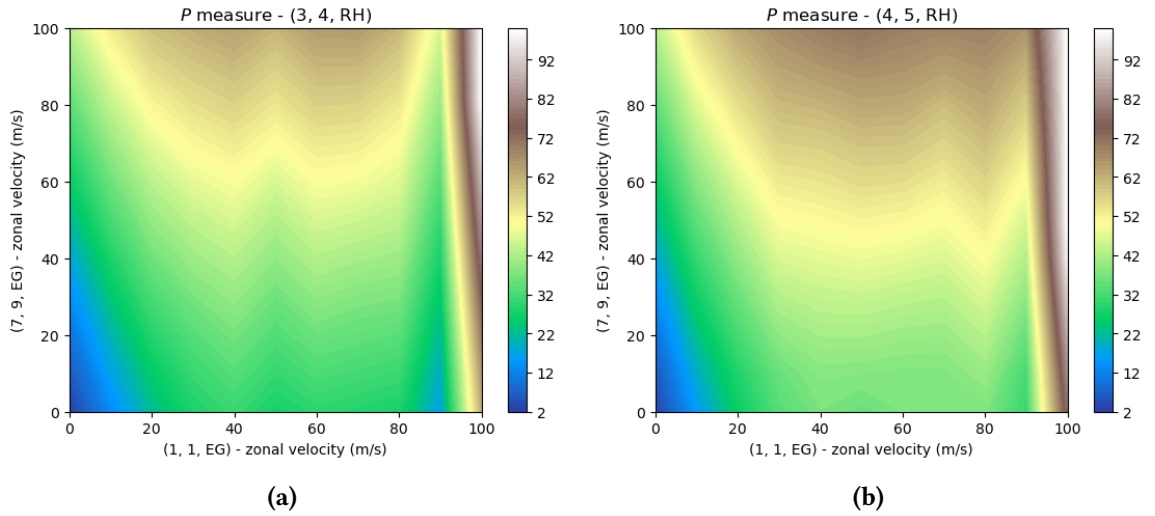


Figure 3.16: $\mathcal{P}$ -measure (%) for the RH modes (3,4) and (4,5) in the five-wave configuration, where all the RH modes have an initial zonal velocity of 30m/s . Here, a grid of 50×50 points was considered.

The dominant periods obtained by Fourier analysis appear to depend only on the initial velocity of the EG mode (1,1), as considered in Figure 3.17. This can be a consequence of the fact that variations in the zonal velocity of the EG mode (7,9) do not alter the periods of the triad containing only RH modes, and consequently Figure 3.17a resembles Figure 3.12a when the RH mode (1,2) has a fixed zonal velocity at 30m/s .

As a final remark, slight changes in the initial conditions of the five-wave system (3.7) - (3.11) can significantly alter the dynamics undergone by the modes. In particular, changing

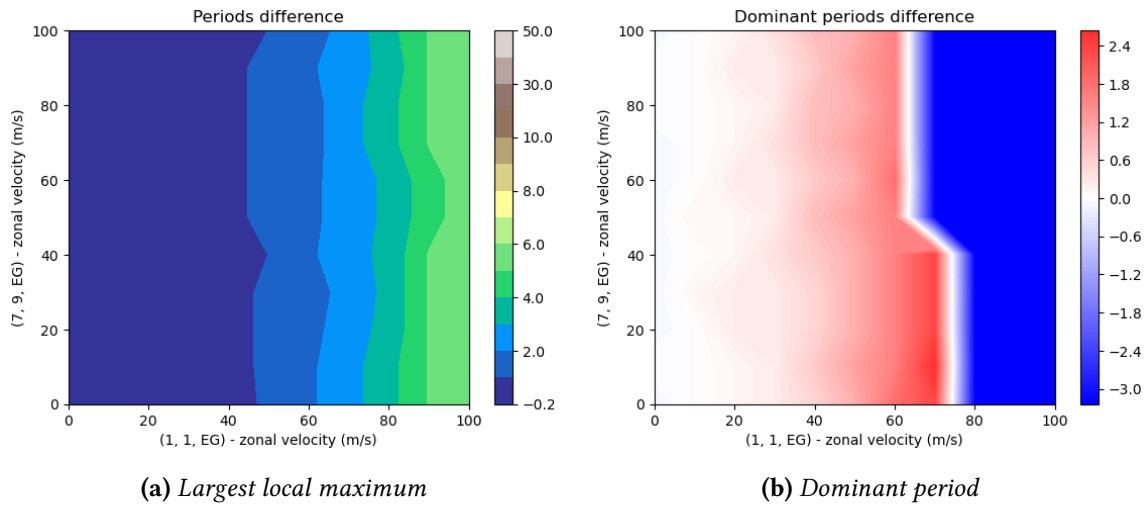


Figure 3.17: (a) The difference in days between the largest-period local maximum in the Fourier spectrum for the five-wave configuration and the triad of RH modes. (b) The difference in days between the dominant periods. Here, a grid of 50×50 points was considered.

the initial zonal velocity of the RH mode (1,2) from 30m/s to 36m/s, which corresponds to the lower white diamond in Figure 3.12a, the periods obtained for the five-wave dynamic are presented in Figure 3.18, where regions characterized by periods of more than 50 days again appear. The structure is similar to the case where the EG mode (7,9) is not present, even though this mode increases the total energy of the system and therefore, the interactions are stronger with more energy being exchanged among the waves.

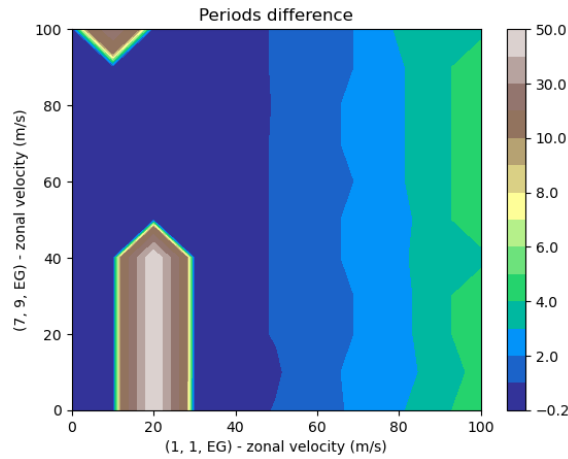


Figure 3.18: The difference in days between the largest local maximum in the Fourier spectrum for the five-wave configuration and the triad of RH modes, where the RH mode (3,4) has a initial zonal velocity of 36m/s.

Chapter 4

Conclusions

The nonlinear interactions between fast and slow waves are related to important phenomena in the atmosphere, such as the Madden-Julian Oscillation (MJO), and due to potential difficulties of numerically representing the gravity (fast) waves and their effects in weather forecasting models, as the latter waves are commonly damped or filtered out from the numerical models. In this work, using a Rotating Shallow Water (RSW) model on the sphere, the nonlinear interactions between fast (gravity) and slow (Rossby-Haurwitz) waves were investigated.

First, a reduced system was used to analyze the nonlinear interactions considering the energy exchanges in a triad of normal modes. In this model, two RH modes can excite a gravity wave initialized with zero kinetic energy, and the opposite is also observed, i.e., two gravity waves can generate an energy flow towards a RH mode with zero initial velocity. This suggests that relevant interactions, which alter the energy field of RH waves, are possibly being neglected when gravity waves are not considered in the model. Furthermore, using the analytic solution of the three-wave system, expressions for the energy efficiency of three resonant modes are derived for a special case where the Hamiltonian is zero. In particular, considering that the coupling coefficient for an RH mode in a triad with the other two gravity modes is low, these formulas show that for resonant triads, the gravity waves exchange energy between themselves, despite the RH mode, resembling the harmonic pendulum. Besides that, the analytic solutions enable the characterization of the *pump mode* in terms of the coupling coefficients, concluding particular systems where there is no pump mode and the waves always present the same phase.

The role of a gravity wave in changing the dynamics of RH modes is more evident in a system with four and five waves, where two (or three) triads are coupled. Although these systems do not conserve the total energy derived from the RSW model, they can be used to analyze the kinetic energy of the modes along with the energy exchanges. The kinetic energy centered in an RH mode can increase significantly due to the presence of a gravity wave, as expected, since a larger amount of energy is available in the system. However, even though the fast waves inject more energy into the system, the energy centered in the RH modes can decrease, and the gravity wave inhibits the nonlinear interactions. Furthermore, the gravity waves generate periods of energy exchange that were not present

in the triadic regime, where periods longer than 50 days appear in the dynamics. The dominant period, i.e., the period with the highest Fourier amplitude, is also changed by the presence of the gravity wave, being displaced by 3 days for some initial conditions.

Therefore, using the RSW model, it was possible to show that gravity waves, which are commonly filtered out in weather forecast models, can excite slower waves, changing their energy fields and periods of energy exchange.

In future works, the complete PDE system, using a spectral method, can be considered to analyze the hypothesis raised here about how a gravity wave can change the nonlinear interactions of RH modes. Furthermore, due to the problem of non-conservation of energy encountered in the four- and five-wave configurations, the search for robust and consistent systems, along with their characterization, can be followed. Finally, noting that the strength of the interactions depends on which waves are coupled and also on their initial amplitudes, the *precession resonance mechanism* (cf. [M. D. BUSTAMANTE *et al.*, 2014](#), [RAPHAELINI *et al.*, 2022](#)) can be studied in the context of the RSW equations to find conditions that ensure a greater energy exchange between gravity and RH waves.

Appendix A

Energy conservation

Although the total energy for the (dimensionless) RSW on the sphere is given by

$$E_T = \frac{1}{4\pi} \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} (h|\vec{u}|^2 + (u^2 + v^2 + h^2)) \cos \phi \, d\phi d\lambda$$

a general Galerking expansion (2.5) does not necessarily lead to a system that conserves this energy. In particular, considering the four-wave ansatz

$$\begin{aligned} \vec{u}(\lambda, \phi, t) = & A_a(t)\Theta_a(t)e^{i(\lambda m_a - \omega_a t)} + A_b(t)\Theta_b(t)e^{i(\lambda m_b - \omega_b t)} \\ & + A_c(t)\Theta_c(t)e^{i(\lambda m_c - \omega_c t)} + A_d(t)\Theta_d(t)e^{i(\lambda m_d - \omega_d t)} + c.c. \end{aligned}$$

where the zonal wavenumbers satisfy the relations

$$\begin{aligned} m_a &= m_b + m_c \\ m_a &= m_b + m_d \end{aligned}$$

the RSW model is reduced to the four-equations system

$$\begin{aligned} \frac{dA_a}{dt} &= -i\omega_a A_a + i\alpha_{b,c}^a A_b A_c + i\alpha_{b,d}^a A_b A_d \\ \frac{dA_b}{dt} &= -i\omega_b A_b + i\alpha_{a,c}^b A_a A_c^* + i\alpha_{a,d}^b A_a A_d^* \\ \frac{dA_c}{dt} &= -i\omega_c A_c + i\alpha_{a,b}^c A_a A_b^* \\ \frac{dA_d}{dt} &= -i\omega_d A_d + i\alpha_{a,b}^d A_a A_b^* \end{aligned}$$

which was used in Chapter 3 to analyze the interactions between a gravity mode and three RH modes.

Using the aforementioned properties of Hough harmonics, the kinetic energy is given simply by

$$E^{(2)}(t) = |A_a|^2 + |A_b|^2 + |A_c|^2 + |A_d|^2$$

while the cubic energy is written as

$$E^{(3)}(t) = 2\text{Re}(A_a^* A_b A_c) S_{a,b,c} + 2\text{Re}(A_a^* A_b A_d) S_{a,b,d}$$

where the term $S_{a,b,c}$ is as in (2.34).

Noting that

$$\begin{aligned} \frac{d}{dt}|A_a|^2 &= \frac{d}{dt}(A_a \cdot A_a^*) = 2\text{Re}\left(\frac{d}{dt}A_a \cdot A_a^*\right) \\ &= 2\text{Re}\left(-i\omega_a|A_a|^2 + i\alpha_{b,c}^a A_a^* A_b A_c + i\alpha_{b,d}^a A_a^* A_b A_d\right) \\ &= -2\alpha_{b,c}^a \mathbb{Im}(A_a^* A_b A_c) - 2\alpha_{b,d}^a \mathbb{Im}(A_a^* A_b A_d) \\ \frac{d}{dt}|A_b|^2 &= 2\text{Re}\left(-i\omega_b|A_b|^2 + i\alpha_{a,c}^b A_a A_b^* A_c^* + i\alpha_{a,d}^b A_a A_b^* A_d^*\right) \\ &= 2\alpha_{a,c}^b \mathbb{Im}(A_a^* A_b A_c) + 2\alpha_{a,d}^b \mathbb{Im}(A_a^* A_b A_d) \end{aligned}$$

and similarly for modes c and d , the derivative of the kinetic energy is written as

$$\frac{d}{dt}E^{(2)} = 2\mathbb{Im}(A_a^* A_b A_c)(\alpha_{a,c}^b + \alpha_{a,b}^c - \alpha_{b,c}^a) + 2\mathbb{Im}(A_a^* A_b A_d)(\alpha_{a,d}^b + \alpha_{a,b}^d - \alpha_{b,d}^a)$$

Furthermore, since

$$\begin{aligned} \frac{d}{dt}(2\text{Re}(A_a^* A_b A_c)) &= 2\text{Re}\left(\frac{d}{dt}A_a^* A_b A_c\right) \\ &= 2\text{Re}\left(i\omega_a A_a^* A_b A_c - i\alpha_{b,c}^a |A_b|^2 |A_c|^2 - i\alpha_{b,d}^a |A_b|^2 A_c A_d^* \right. \\ &\quad \left. - i\omega_b A_a^* A_b A_c + i\alpha_{a,c}^b |A_a|^2 |A_c|^2 + i\alpha_{a,d}^b |A_a|^2 A_c A_d^* \right. \\ &\quad \left. - i\omega_c A_a^* A_b A_c + i\alpha_{a,b}^c |A_a|^2 |A_b|^2\right) \\ &= 2\delta_{a,b,c} \mathbb{Im}(A_a^* A_b A_c) + 2(\alpha_{b,d}^a |A_b|^2 - \alpha_{a,d}^b |A_a|^2) \mathbb{Im}(A_c A_d^*) \end{aligned}$$

and similarly for $2\text{Re}(A_a^* A_b A_d)$, the derivative of the cubic energy is given by

$$\begin{aligned} \frac{d}{dt}E^{(3)} &= 2\delta_{a,b,c} \mathbb{Im}(A_a^* A_b A_c) S_{a,b,c} + 2\delta_{a,b,d} \mathbb{Im}(A_a^* A_b A_d) S_{a,b,d} \\ &\quad + 2(\alpha_{b,d}^a |A_b|^2 - \alpha_{a,d}^b |A_a|^2) \mathbb{Im}(A_c A_d^*) S_{a,b,c} \\ &\quad + 2(\alpha_{b,c}^a |A_b|^2 - \alpha_{a,c}^b |A_a|^2) \mathbb{Im}(A_d A_c^*) S_{a,b,d} \end{aligned}$$

Therefore, the derivative of the total energy is then

$$\begin{aligned} \frac{d}{dt}(E^{(2)} + E^{(3)}) &= 2\mathbb{Im}(A_a^* A_b A_c) (\alpha_{a,c}^b + \alpha_{a,b}^c - \alpha_{b,c}^a + \delta_{a,b,c} S_{a,b,c}) \\ &\quad + 2\mathbb{Im}(A_a^* A_b A_d) (\alpha_{a,d}^b + \alpha_{a,b}^d - \alpha_{b,d}^a + \delta_{a,b,d} S_{a,b,d}) \\ &\quad + 2(\alpha_{b,d}^a |A_b|^2 - \alpha_{a,d}^b |A_a|^2) \mathbb{Im}(A_c A_d^*) S_{a,b,c} \\ &\quad - 2(\alpha_{b,c}^a |A_b|^2 - \alpha_{a,c}^b |A_a|^2) \mathbb{Im}(A_d A_c^*) S_{a,b,d} \end{aligned}$$

Provided that the coupling coefficients and the integral terms $S_{a,b,c}$ and $S_{a,b,d}$ are the *same* as for the isolated triads, the terms in blue vanish, in such a manner that the variation on the total energy is

$$\begin{aligned}
 \frac{d}{dt} (E^{(2)} + E^{(3)}) &= 2(\alpha_{b,d}^a |A_b|^2 - \alpha_{a,d}^b |A_a|^2) \mathbb{I}m(A_c A_d^*) S_{a,b,c} \\
 &\quad - 2(\alpha_{b,c}^a |A_b|^2 - \alpha_{a,c}^b |A_a|^2) \mathbb{I}m(A_c A_d^*) S_{a,b,d} \\
 \frac{d}{dt} (E^{(2)} + E^{(3)}) &= 2(\alpha_{a,c}^b S_{a,b,d} - \alpha_{a,d}^b S_{a,b,c}) |A_a|^2 \mathbb{I}m(A_c A_d^*) \\
 &\quad + 2(\alpha_{b,d}^a S_{a,b,c} - \alpha_{b,c}^a S_{a,b,d}) |A_b|^2 \mathbb{I}m(A_c A_d^*)
 \end{aligned} \tag{A.1}$$

The energy should be conserved for any initial condition, and consequently it is conserved if, and only if

$$\alpha_{a,c}^b S_{a,b,d} - \alpha_{a,d}^b S_{a,b,c} = 0$$

which indicates that the coupling coefficients of the two triads belonging to the four-wave configurations must satisfy

$$\frac{\alpha_{a,c}^b}{S_{a,b,c}} = \frac{\alpha_{a,d}^b}{S_{a,b,d}} \tag{A.2}$$

a relation that *does not hold* in general.

By (A.1) the variation of the total energy depends on the product of *four* amplitudes, and consequently, for smaller amplitudes, the energy can be thought of as constant. This was the case considered in Chapter 3, where the initial amplitudes were chosen in the order of 10^{-1} , leading to a variation of less than 10^{-4} , which in terms of total energy represents less than 2%. However, considering higher initial velocities the derivative increases and the conservation can no longer be used.

Recalling one of the examples presented in Chapter 3 (see Table A.1), where the four-wave configuration leads to the system here considered, Table A.2 shows that the condition for conservation (A.2) is not satisfied and the total energy varies following the derivative expressed in (A.1).

		Freq.	Coeff. 1	Coeff. 2
Mode a	4,5,RH	-0.12522	0.3638	-0.5572
Mode b	3,4,RH	-0.13431	0.4434	-0.5902
Mode c	1,2,RH	-0.09934	-0.0284	-
Mode d	1,1,EG	0.36913	-	-0.1323

Table A.1: Four-wave configuration to study the non-conservation of energy.

Initializing the above system of waves with all RH modes having zonal velocity $30m/s$ and the EG mode $80m/s$ (see Table 3.3, example 2), the energies undergo the dynamics presented in Figure A.1a, where the energies were normalized by a division by $E_T(0)$. The derivative of the total energy is displayed in Figure A.1b, where the orange line is the

		$S_{a,b,c}$	$S_{a,b,d}$	$\alpha_{c}^- / S_{a,b,c}$	$\alpha_{d}^- / S_{a,b,d}$
Mode a	4,5,RH	0.471053	0.459251	0.772484	-1.21325
Mode b	3,4,RH	0.471053	0.459251	0.9412938	-1.28516

Table A.2: Relations of conservation for the quartet in Table A.1.

analytic derivative (A.1) and the blue dots were obtained by a numerical approximation of the derivative using the fraction

$$E'_T(t) \approx \frac{E_T(t+h) - E_T(t)}{h}$$

with $h = 0.01$.

One notes that the derivative is $O(10^{-4})$ as expected and the numerical approximation corresponds satisfactorily to the analytic formula obtained. The kinetic and cubic energies are in anti-phase with almost the same amplitude, which also contributes to the low derivative.

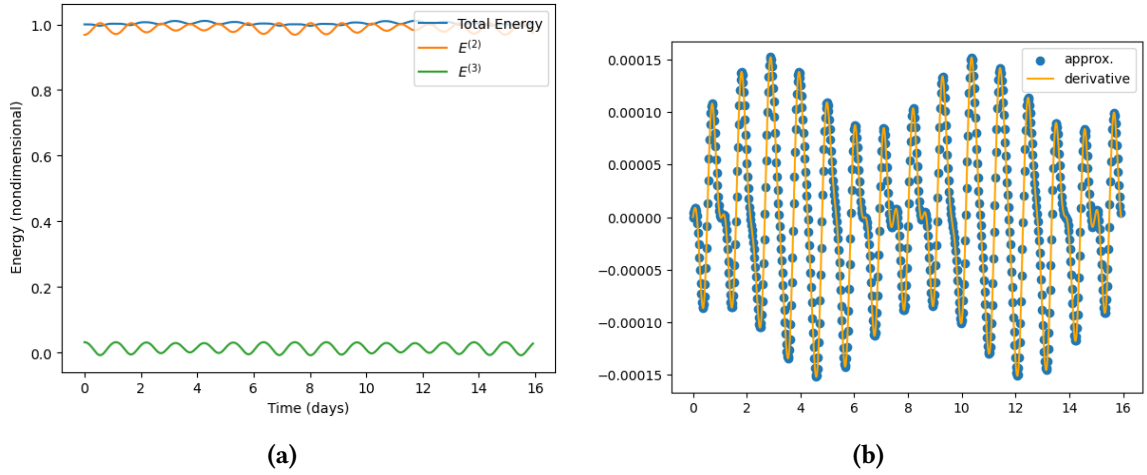


Figure A.1: (a) Total, kinetic and cubic energy for the quartet in Table 3.3 example 2. (b) The analytic and numeric derivatives of the total energy.

However, changing the initial amplitudes to 1 for all modes, corresponding to zonal velocities larger than $300m/s$ the energies undergo dynamics as in Figure A.2a, where the total energy is dominated by the cubic energy. Figure A.2b shows the derivative of the total energy, which presents higher values, guaranteeing its evident non conservation.

Therefore, depending on the Galerking expansion, the system derived from the RSW equations may not conserve the associated energy. The same analysis can be made for systems of 5 waves, leading to the same results of non-conservation.

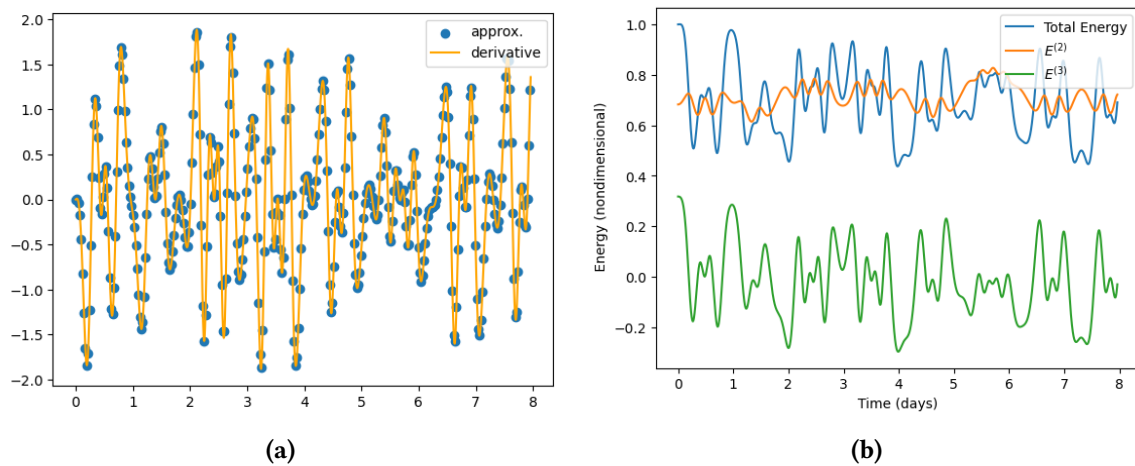


Figure A.2: (a) Total, kinetic and cubic energy for the quartet in Table 3.2 where all the amplitudes are 1. (b) The analytic and numeric derivatives of the total energy.

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